



PE CIVIL-STRUCTURAL EXAM 02

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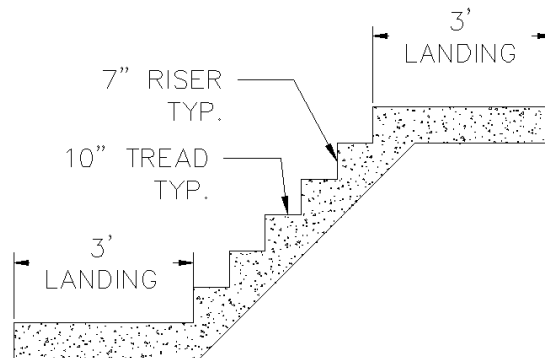
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Problems

[Click here for Solutions](#)

1. Which of the following is not true about dead loads on the roof of a building?
 - (A) Dead loads consist of the weight of materials of construction incorporated in the building roof including insulation layers,
 - (B) Installed solar panels and their support systems including framing and ballast are all considered dead loads,
 - (C) Weight of soil layers in landscaped roofs should be considered as dead loads, and the moisture content should be taken as the average of dry and fully saturated conditions.
 - (D) Dead loads include fixed service equipment such as cranes and material handling system

2. The flight of stairs shown below belongs to a school building. The width of the stairs and landing is 4 ft. Calculate the total live load following the provisions of International Building Code.

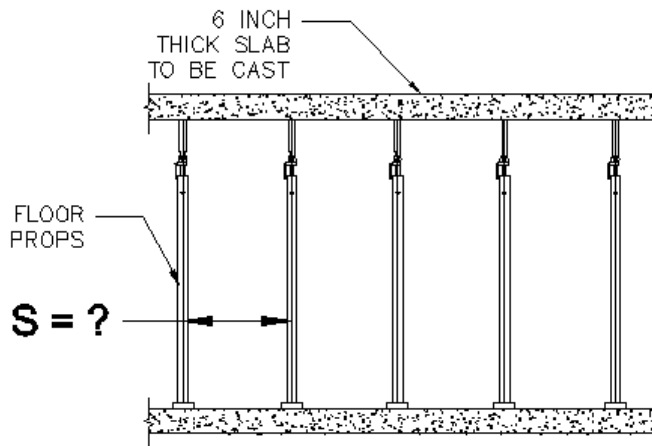


- (A) 1626 lb
- (B) 2601 lb
- (C) 4067 lb
- (D) 2000 lb

3. A two-story gymnasium is to be built in New York. The ground floor slab is on grade and the first floor will be supported by steel beams. The tributary area for an interior beam of length 40 ft is 1250ft^2 . What is the minimum live load acting on the interior beam? Apply reduction factors as applicable.

(A) 6.25 kips/ft
(B) 3.125 kips/ft
(C) 2.5 kips/ft
(D) 1.5 kips/ft

4. A 6-inch thick fresh concrete slab is to be cast on a new floor. The construction load is estimated as 50 psf including equipment and weight of formwork. The newly cast floor is to be shored with floor props that have allowable carrying capacity 2 kips. The props are to be laid out following a square layout, with equal spacing in both directions. What is the maximum permitted spacing? Unit weight of concrete = 150 pcf



(A) 2ft
(B) 3ft
(C) 4ft
(D) 5ft

5. An 80 ft tall commercial building is constructed in a river valley located 2000 ft above sea level. The basic wind speed in the area is 90 mph. The building is classified as Risk category III and lies in the Exposure B category terrain. The building can be considered enclosed. Topographic effects in the area can be neglected. Calculate the velocity pressure acting on the MWFRS at the top of the building. The structure can be considered rigid.

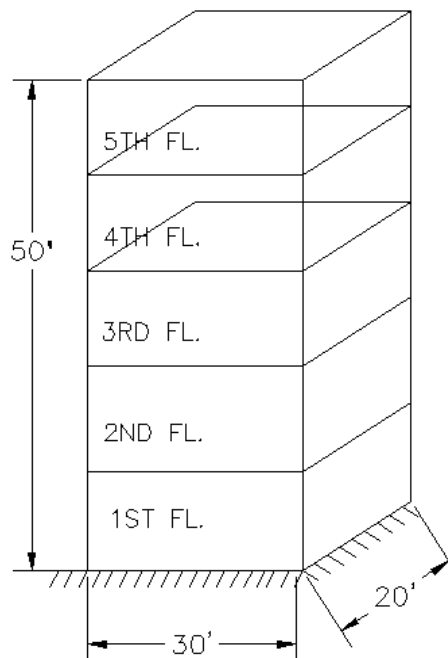
(A) 11 lb/ft²
(B) 12.81 lb/ft²
(C) 15.24 lb/ft²
(D) 22.3 lb/ft²

6. A 10 story building is identified as a critical infrastructure and is assigned risk category IV. The building is located in an active seismic zone with $S_s = 1g$ and $S_1 = 0.4g$. The bearing stratum is explored to consist of stiff soil. Determine seismic design category of the structure following procedures of ASCE 7-16.

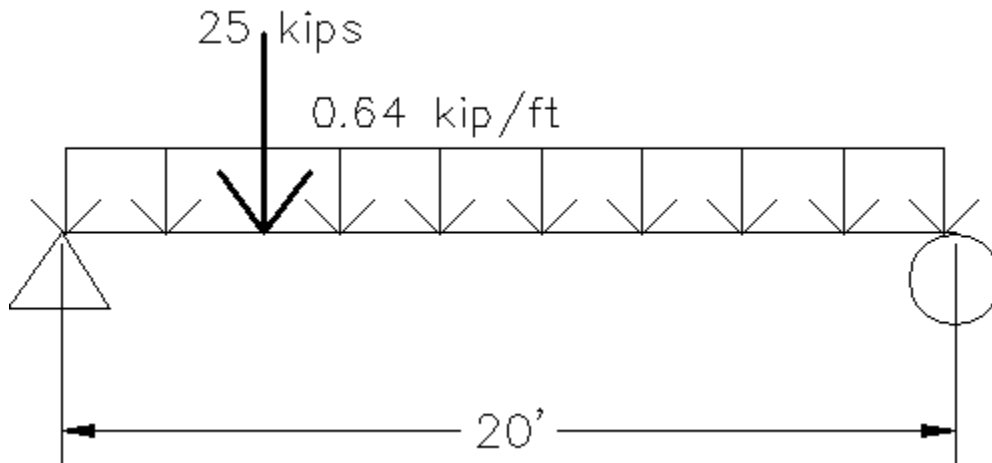
(A) Seismic Design Category F
(B) Seismic Design Category A
(C) Seismic Design Category C
(D) Seismic Design Category D

7. The dimensions of a 5 story building are as shown in the sketch below. The building is made of reinforced concrete moment resisting frames in which the frames resist all lateral and gravity loads. What is the estimated fundamental time period of the building?

(A) 0.45 seconds
(B) 0.5 seconds
(C) 0.55 seconds
(D) 0.6 seconds



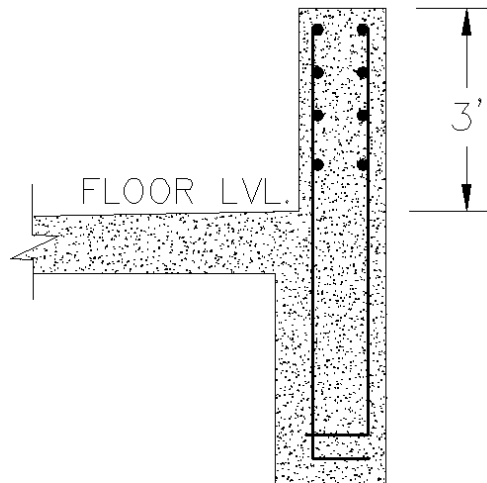
8. A single span simply supported bridge is subjected to a uniform lane load of 0.64 kip/ft and a moving concentrated wheel load of 25 kips. What is the maximum reaction acting on the support of the bridge?



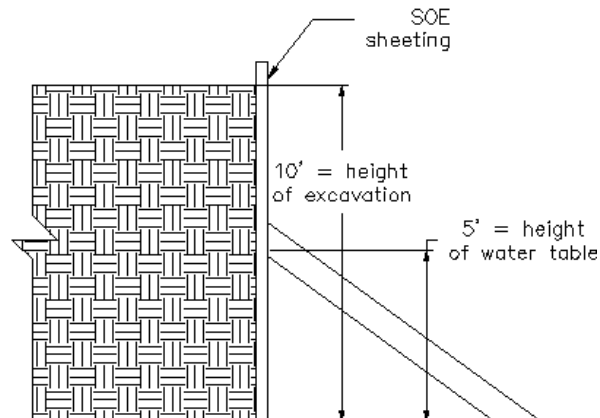
- (A) 18.9 kips
(B) 25 kips
(C) 31.4 kips
(D) 37.8 kips
9. Which of the following is incorrect about rain on snow surcharge load per ASCE 7-16?
- (A) Rain on surcharge load is applied in addition to uniform roof snow load for balanced loading condition,
(B) For an area with ground snow load 18 psf, rain surcharge may apply,
(C) Rain surcharge for a roof with slope 20 degrees and horizontal distance from eave to ridge 20ft is 5 psf,
(D) For a roof with slope 0.5 degrees and horizontal distance from eave to ridge is 30ft, if the uniform roof snow load is 16 psf, the total load including rain surcharge is 21psf

10. Given detail below is the section of a barrier in a 2nd floor parking lot. The barrier system is designed per the provisions of ASCE 7-16 for impact by passenger vehicles. What is the maximum factored design moment acting at the slab level?

- (A) 9 kip*ft
(B) 13.5 kip*ft
(C) 14.4 kip*ft
(D) 21.6 kip*ft



11. What is the total horizontal force acting per unit length of the SOE sheeting as shown in the figure below? The height of excavation is 10ft and the water table is at 5ft. The active earth pressure coefficient is 0.33. The dry density of the soil is 110 pcf and the submerged density is 65 pcf.

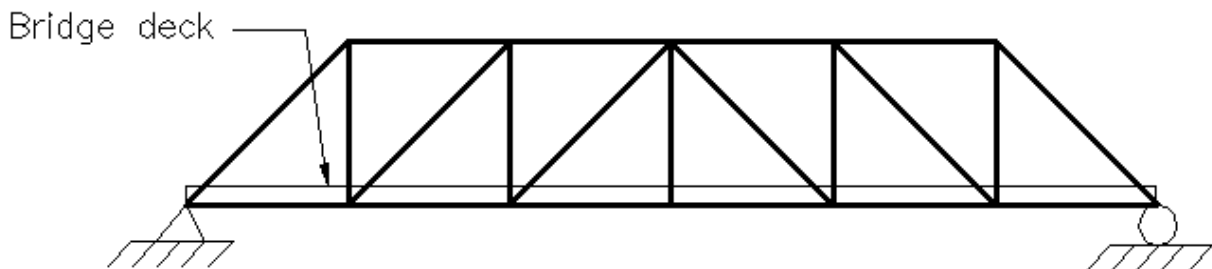


- (A) 1.1 kips/ft
(B) 1.3 kips/ft
(C) 1.9 kips/ft
(D) 2.4 kips/ft

12. Which of the following is not correct about lateral earth pressure?

- (A) Lateral earth pressure coefficient is the ratio of horizontal stress to vertical stress exerted by the soil,
- (B) The strain required to achieve passive pressure is significantly larger than the strain required to reach active pressure,
- (C) Active earth pressure cannot be negative regardless of properties of soil,
- (D) The lateral pressure exerted by soil retained by bridge abutments connected to the superstructure deck above is calculated by using the coefficient of earth pressure at rest.

13. The elevation of a single span simply supported truss bridge is shown below. When a car passes through the deck, what is the correct distribution of forces in the members of the truss? Assume lateral loads are not acting.



- (A) Diagonals in Tension, vertical members in compression, upper chord in compression and bottom chord in tension,
- (B) Diagonals in Compression, vertical members in tension, upper chord in compression and bottom chord in tension,
- (C) Diagonals in compression, vertical members in tension, upper chord in tension and bottom chord in compression,
- (D) Diagonals in Tension, vertical members in compression, upper chord in tension and bottom chord in compression.

14. The load effects on the main girder of a single lane bridge are as follows:

- a. Service dead load moment = 800 kip*ft
- b. Service live load moment due to design truck = 250 kip*ft
- c. Service live load moment due to lane load = 300 kip*ft

Force effects due to temperature, creep, settlement, water load and friction can be neglected,

What is the maximum factored moment acting on the girder per Strength I combination?

- (A) 2150 kip*ft
- (B) 2328 kip*ft
- (C) 2785 kip*ft
- (D) 2966 kip*ft

15. The cross-sectional areas of each of the shapes shown below are equal.



OPTION A



OPTION B



OPTION C



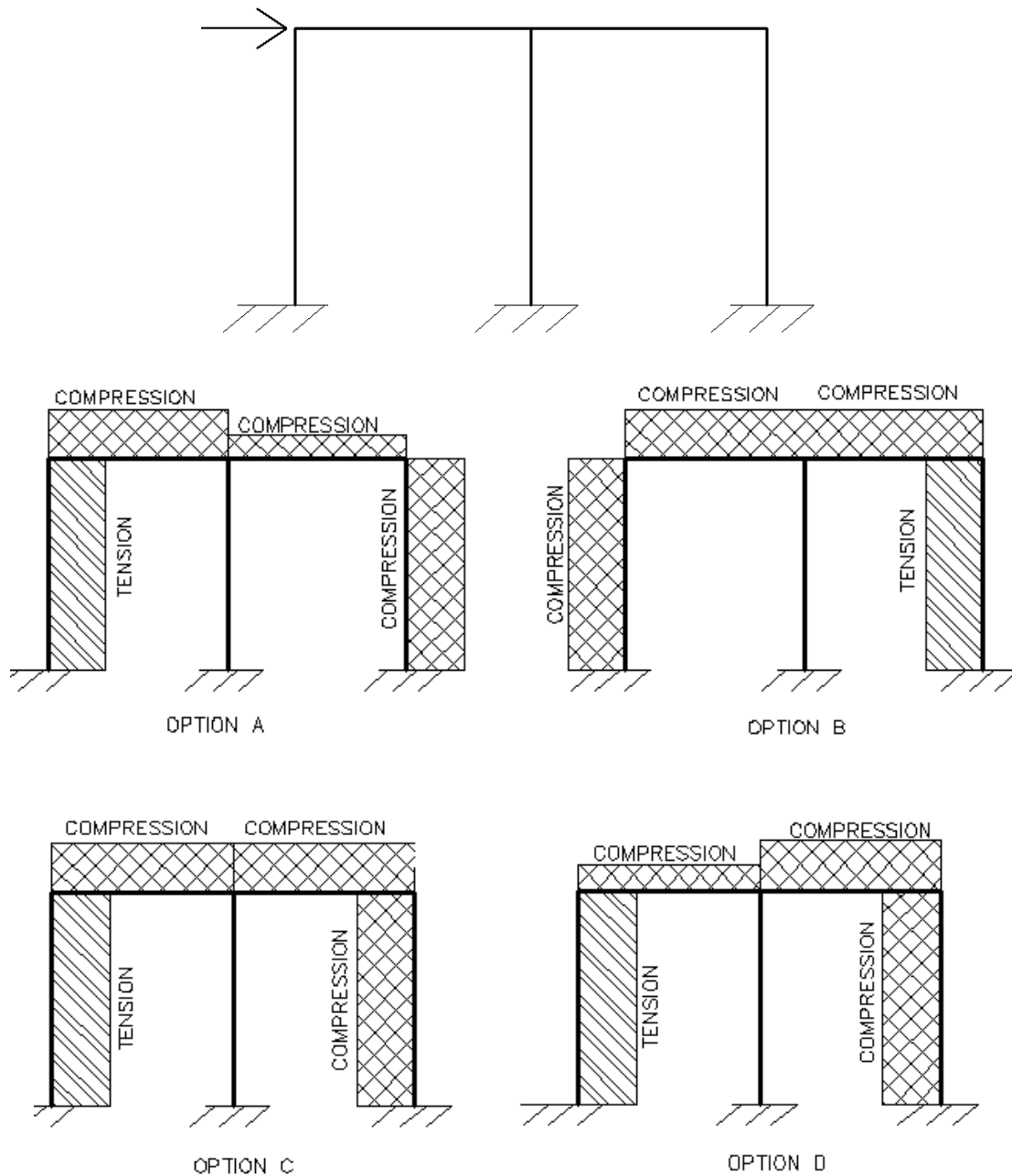
OPTION D

(X-SECTIONS ARE NOT TO SCALE)

Which shape is best suited for torsional applications?

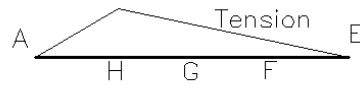
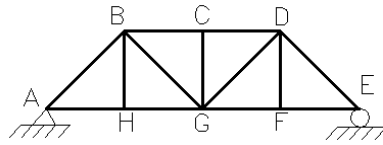
- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

16. The given multi bay frame has a lateral load acting on the top. Select the correct axial force diagram.

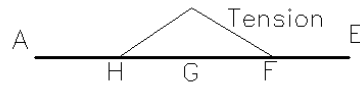


- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

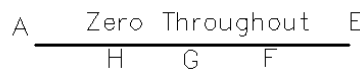
17. For the truss shown below, select the correct influence line diagram for member CG when a point load passes through A to E



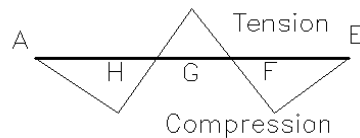
OPTION A



OPTION B



OPTION C



OPTION D

- (A) Option A
(B) Option B
(C) Option C
(D) Option D

18. A 30 ft long member is subjected to tensile stresses and temperature strains. The material is linearly elastic with $E = 29000$ ksi and is subjected to a tensile load of 200 kips. The length of the material was measured at 32°F . What is the length of the member under the present loading condition at a temperature of 100°F ?

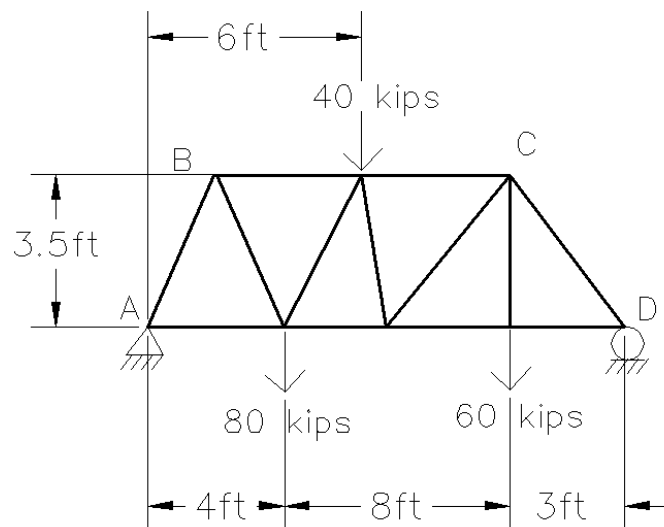
Coefficient of thermal expansion can be taken as $0.000007 / ^\circ\text{F}$. The Cross section Area of the member is 20 in^2 .

- (A) 0.124 in
(B) 0.171 in
(C) 0.295 in
(D) 0.51 in

19. A 30 ft long beam with cross sectional area 15 in^2 is shown below. The beam is fixed on one end and has a 0.1 inch clearance to the support at the other end to allow for thermal expansion. The beam was installed when the ambient temperature was 30°F . The material is linearly elastic with $E = 29000 \text{ ksi}$ and not subjected to any load. The 30ft length of the beam was measured at 30°F . What is the axial load in the member during summer at a temperature of 100°F ? Coefficient of thermal expansion can be taken as $0.000007 / ^\circ \text{F}$.

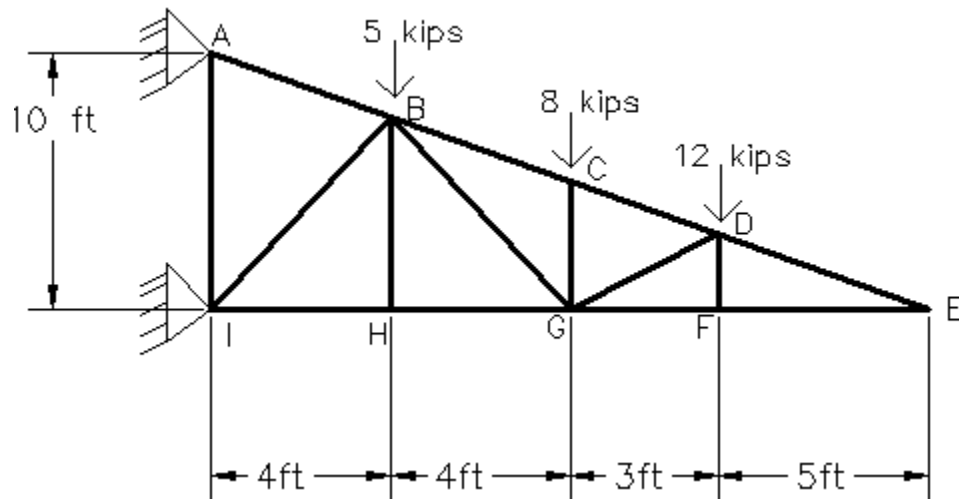


- (A) 0 kips
 (B) 92.3 kips
 (C) 158.3 kips
 (D) 213.15 kips
20. For the simply supported truss shown below, calculate the force acting in member CD.

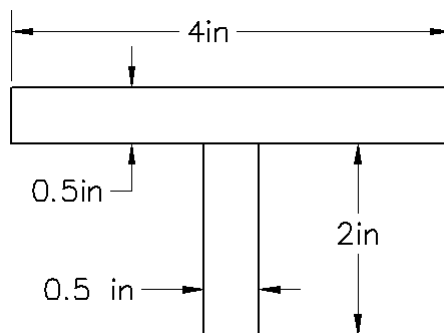


- (A) 112 kips (compression)
 (B) 112 kips (tension)
 (C) 131 kips (compression)
 (D) 131 kips (tension)

21. For the truss shown below, calculate the forces in members BH and DF.

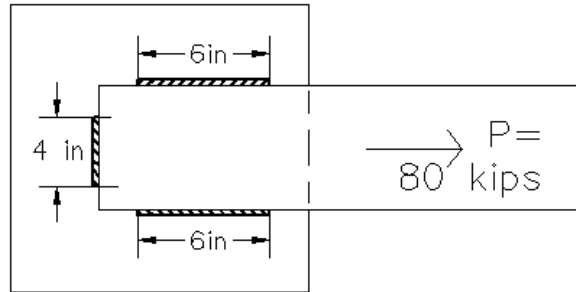


- (A) BH = 0 DF = 12 kips
 (B) BH = 0 DF = 0
 (C) BH = 5 kips DF = 12 kips
 (D) BH = 5 kips DF = 0
22. The maximum shear force acting on the T beam section shown below is 2.5 kips. What is the shear stress at the junction of the flange and web?



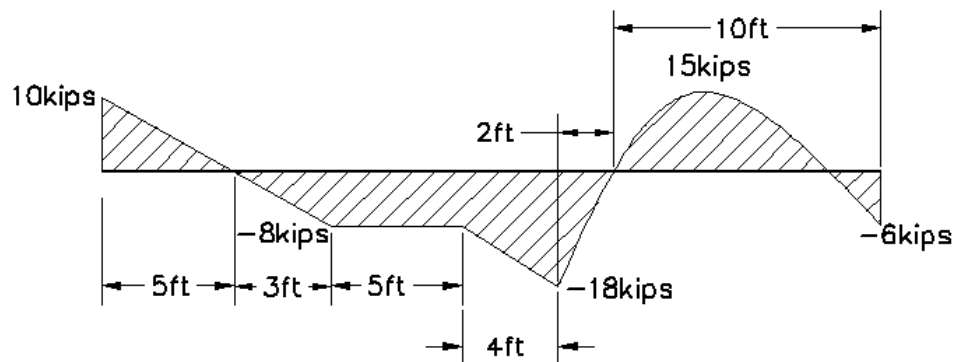
- (A) 1313 psi
 (B) 1978 psi
 (C) 2618 psi
 (D) 2943 psi

23. Two plates are welded as shown in the plan below. A load of 80 kips acts in the smaller plate concentric to the centroid of the weld group. What is the shear acting on the welds?



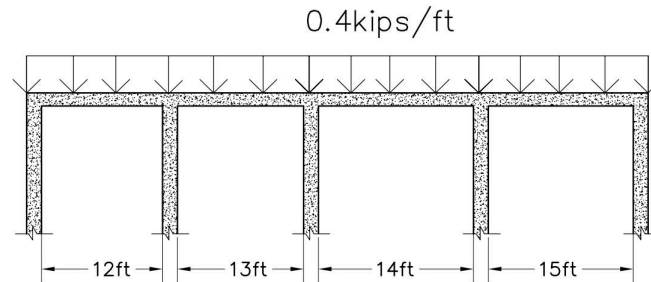
- (A) 3 kip/in
- (B) 4 kip/in
- (C) 4.5 kip/in
- (D) 5 kip/in

24. The shear force diagram for a simply supported beam is given below. What is the value of the bending moment at a section 11 ft from the left support?

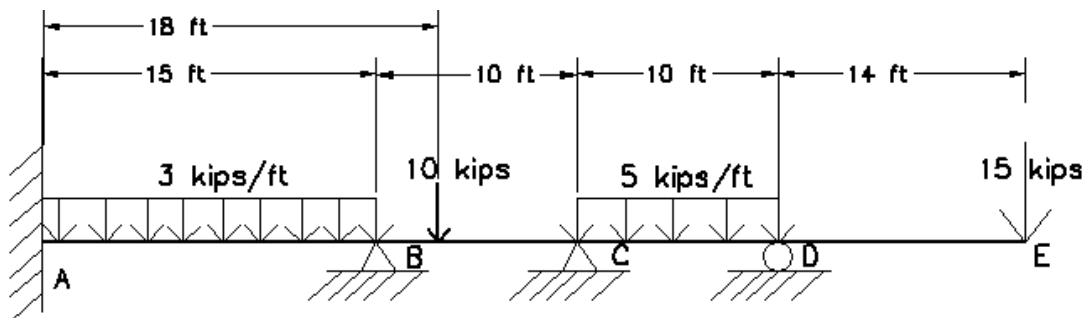


- (A) 9 kip*ft
- (B) -9 kip*ft
- (C) -11 kip*ft
- (D) 13 kip*ft

25. The figure below shows a monolithically constructed reinforced concrete beam and column frame. The factored load acting on the beam is 0.4 kips/ft. What is the approximate maximum positive and negative moment acting on the concrete beams? Assume all conditions for ACI's simplified analysis are satisfied.



- (A) +3.4 kip*ft , -6.8 kip*ft
 (B) + 4.7 kip*ft , -6.6 kip*ft
 (C) + 6.4 kip*ft , -8.4 kip*ft
 (D) + 8.2 kip*ft , -11.8 kip*ft
26. For the continuous beam shown below, using the moment distribution method, what is the unbalanced portion of fixed end moments for joint C?

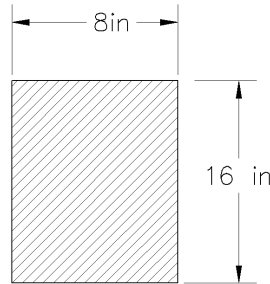


- (A) 23.2 kip*ft
 (B) 31.43 kip*ft
 (C) 35.37 kip*ft
 (D) 38.22 kip*ft

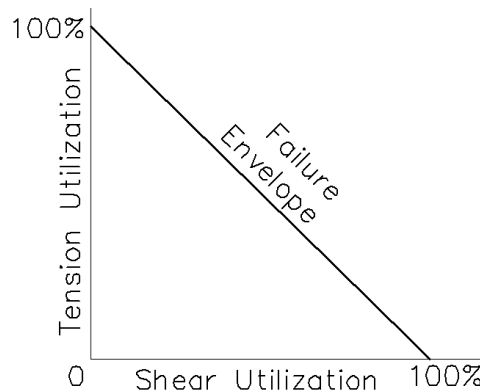
27. A rectangular beam is subjected to a compressive load of 150 kips. In addition to the compressive force it is subjected to biaxial bending. The dimensions of the beam is 8 in X 16 in. What is the maximum normal stress acting on the beam section?

Moment about stronger axis = 100 kip*ft

Moment about weaker axis = 50 kip*ft

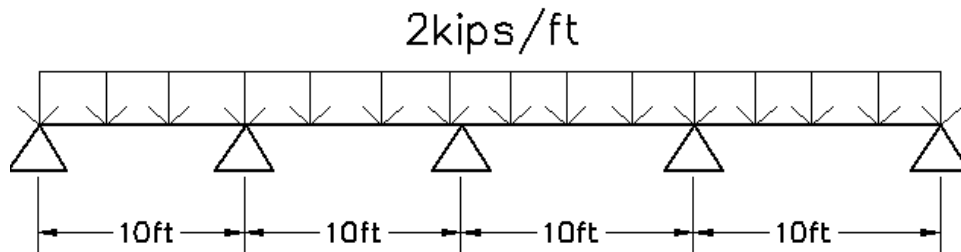


- (A) 1.2 ksi
(B) 5.6 ksi
(C) 8.2 ksi
(D) 12.1 ksi
28. For a section subjected to combined tension and shear, the failure envelope is given by the linear graph below. What is not true about the failure due to interaction of combined stresses?

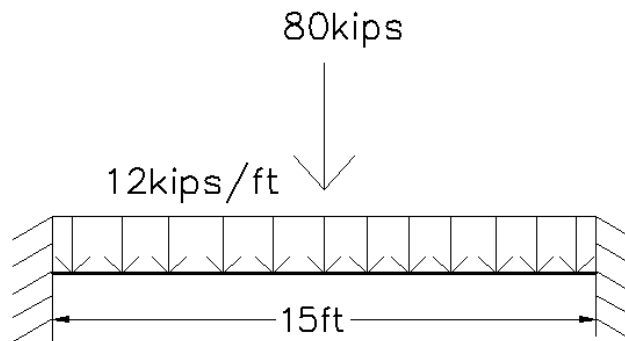


- (A) The section can have residual tension capacity and residual shear capacity but still fail in interaction,
(B) The section can have no tension load but still fail when interaction is considered,
(C) The section can have no shear load but still fail when interaction is considered,
(D) If the section is utilized 60% for tension and 50% for shear, then it passes the design.

29. A four span W10X45 steel beam has a uniformly distributed load of 2 kips/ft as shown below. What is the maximum deflection in the beam? The modulus of elasticity of steel $E = 29000$ ksi. The bending moment acts about the weaker axis of the beam.



- (A) 0.1 in
 (B) 0.15 in
 (C) 0.2 in
 (D) 0.25 in
30. A fixed-fixed beam is subjected to a uniformly distributed load and a point load as shown below. The moment of inertia of the beam section is 84 in^4 and the modulus of elasticity is 29000 ksi. What is the maximum deflection in the beam due to the combined loading?



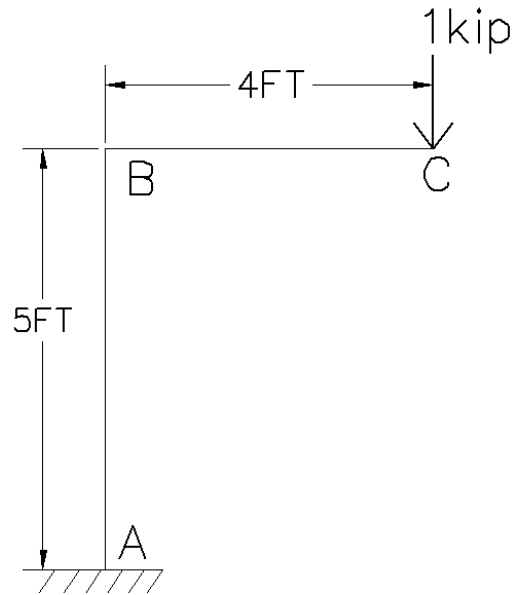
- (A) 1 in
 (B) 1.12 in
 (C) 1.52 in
 (D) 2.12 in

31. Calculate the maximum downward deflection at point C due to a load of 1 kip acting as shown in the drawing below.

Moment of inertia of column AB = $2I$

Moment of inertia of beam BC = I .

Modulus of elasticity = E .



- (A) $184/3EI$
(B) $284/3EI$
(C) $328/4EI$
(D) $584/5EI$
32. A fixed-fixed beam with a rectangular cross section is designed to support a concentrated load at the center of the span. Due to construction extensions, the beam has to be doubled in length. The new beam will be supporting the same concentrated load at mid span as the original beam. To restrict the deflections to the original configuration, what changes should be made to the new beam?

- (A) Double the depth,
(B) Double the breadth,
(C) Double the width and breadth,
(D) Increase cross section area two times.

33. Which of the following is not true about steel column buckling?

- (A) Inelastic buckling strength is less than elastic buckling strength,
- (B) Euler's buckling equation predicts buckling load which when released, causes the member to return to its undeformed form,
- (C) Buckling deflection is sudden and catastrophic,
- (D) Elastic buckling strength of a material is always less than the yield strength of the material.

34. A steel connection is subjected to a maximum stress range of 10 ksi in its service life. Through experimental testing, the constant amplitude fatigue limit for the detail is determined as 12 ksi. What is the estimated fatigue life of the connection?

- (A) 0,
- (B) 10000,
- (C) Infinite,
- (D) Cannot be determined without the modulus of elasticity.

35. Which of the following is not true about procedures related to special inspections?

- (A) The owner, engineer of record, contractor and building officials have to approve the competency of the party serving as the special inspector,
- (B) For a welding job, in addition to non destructive evaluation reports, the qualifications of the fabricator also needs to be included as a part of inspection documentation,
- (C) Structural observations or special inspections are not mandatory for every project, regardless of the cost of the project,
- (D) The special inspector is not responsible for the contractor's means and methods.

36. Which document governs the content, format, overall procedures and approvals of construction submittals?

- (A) International Building Code ,
- (B) OSHA Technical manual,
- (C) ACI 318-14,
- (D) Contract documents and specifications.

37. A concrete deadman block is used to anchor a cable that provides lateral stability to a falsework system. The maximum force acting on the cables is 5 kips and the horizontal inclination of the cable is 30 degrees. Due to site constraints, the maximum height of the anchor block is limited to 6ft. What is the minimum area of the block that can safely stabilize the system? The cable is attached at the center of the top surface of the block.

Factor of safety = 1.5,

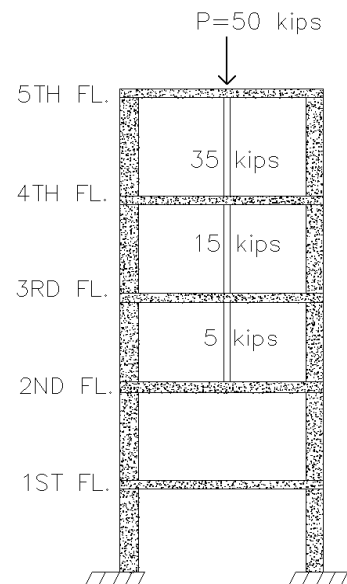
Coefficient of friction between concrete and soil = 0.3,

Density of concrete = 150 lbs/ft³

- (A) 5 ft X 5 ft
- (B) 6 ft X 6 ft
- (C) 7 ft X 7 ft
- (D) 8 ft X 8 ft

38. Three floors are shored to support a concentrated load of 50 kips acting on the fifth floor. The axial loads acting on the 5th floor prop is 35 kips, on the 4th floor prop is 15 kips and on the 3rd floor prop is 5 kips. What is the maximum load distributed to an individual floor slab?

- (A) 15 kips
- (B) 20 kips
- (C) 35 kips
- (D) 50 kips



39. A 10 ft X 5 ft scaffold platform is intended to support a load of 20 psf. The platform is supported by simply supported beams on the two longer sides. Calculate the design moment on the scaffold beam per OSHA regulations. The self weight of the beam is 40 plf.

- (A) 1 kip*ft
- (B) 2 kip*ft
- (C) 3 kip*ft
- (D) 4 kip*ft

40. In a pre-engineered building, a 2nd story bolted rigid moment connection joint has 8 bolts on each side of the web. During erection, what is the minimum number of bolts on each side of the web that needs to be installed and tightened before the hoisting equipment can be safely released?

- (A) 2
- (B) 4
- (C) 6
- (D) 8

41. The particle size distribution of an inorganic soil sample is as follows:

Liquid Limit = 51
Plastic Limit = 19

Determine the USCS classification of the soil sample.

- (A) GC
- (B) SP
- (C) CH
- (D) MH

Soil Size (mm)	Fraction retained on sieve
<0.002	0.3
0.002-0.005	0.12
0.005-0.05	0.11
0.05-0.075	0.16
0.075-1.18	0.31
>1.18	0

42. A soil sample has a density of 1900 kg/m^3 . The water content ratio is 19% and the specific gravity of the sample is 2.7. What is the degree of saturation of the sample?

- (A) 55%
- (B) 65%
- (C) 75%
- (D) 85%

43. Three standard 4 inX 8 in cylinders were sampled from a slab pour job. The samples were cured and tested after 28 days with the following results:

Sample number	Axial compressive load during failure
1	50 kips
2	45 kips
3	57 kips

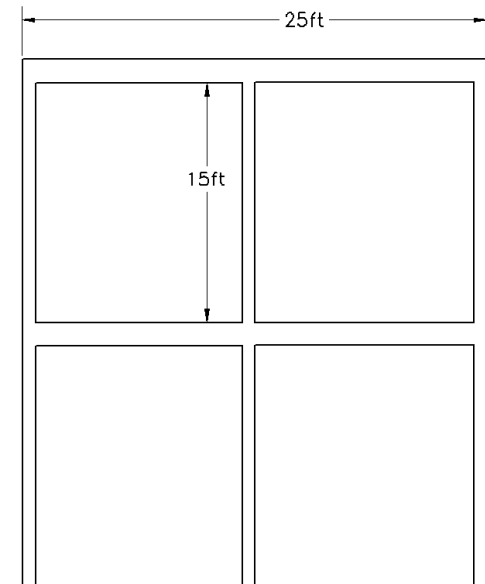
What is the 28 day compressive strength of concrete that is to be used for compliance verification?

- (A) 3581 psi
- (B) 3979 psi
- (C) 4032 psi
- (D) 4536 psi

44. A 12in X 18 in simply supported non prestressed singly reinforced concrete beam section is to be designed with an effective cover of 2 inches. The strength of concrete is 3500 psi and the rebar yield strength is 60 ksi. What is the minimum number of #4 main flexural reinforcement bars that should be used following ACI recommendations?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

45. What is the slenderness ratio of the center-top column shown below per ACI 318-14? The span of each beam is 12.5 ft and height of each column is 15ft. The dimensions of all beams are 12 in X 24 in and the dimensions of all the columns are 14 in X 14 in. The same grade concrete is used for all floors and members.

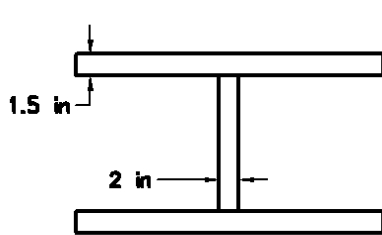


- (A) 22
(B) 31
(C) 39
(D) 47
46. The yield strength of the threaded rod shown below is 120 ksi. The minimum ultimate strength is 150 ksi. What is the minimum allowable tensile strength and the minimum allowable shear strength for bearing type connections using the rod if the material conforms to AISC specifications?

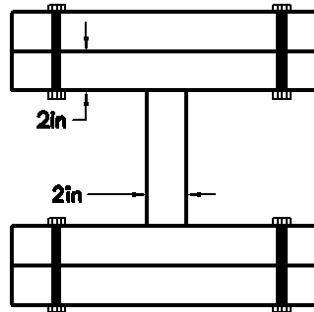


- (A) 56 ksi and 33 ksi
(B) 84 ksi and 50.625 ksi
(C) 90 ksi and 54 ksi
(D) 112.5 ksi and 67.5 ksi

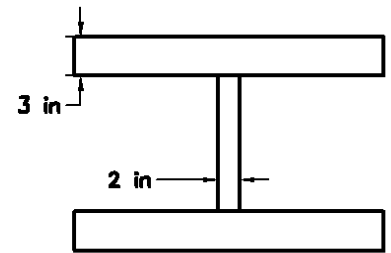
47. Which of the following steel sections require an impact test per AISC specifications?



Section I



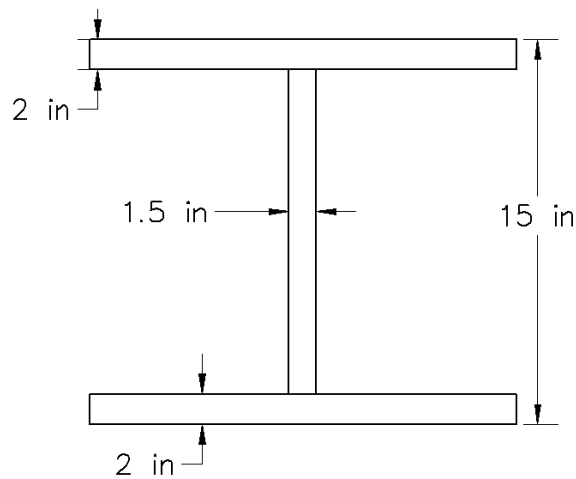
Section II



Section III

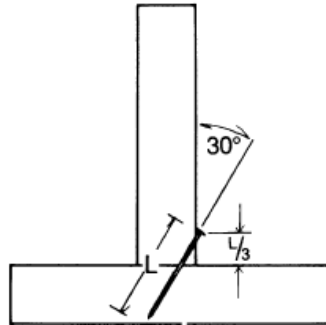
- (A) I, II and III
- (B) I and II only
- (C) I only
- (D) III only

48. A built up plate girder is made using A572 Gr 50 steel plates. The webs of the girder are stiffened with transverse stiffeners with 40 inches clear spacing between the stiffeners. What is the web plate shear buckling coefficient per AISC specifications?

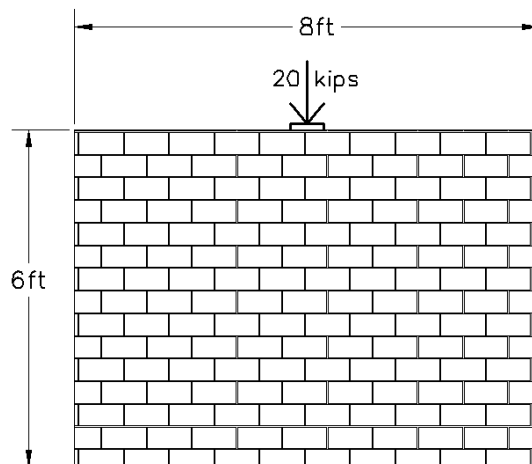


- (A) 5
- (B) 5.34
- (C) 5.53
- (D) 5.7

49. A toe nail connection is made using 3.5 inch long 16D common wire nails between two Douglas Fir Larch members as shown below. The dressed size of the side member is 1.5X1.5. Determine the ASD adjusted lateral load capacity for live loads. Assume dry service conditions during fabrication and service and temperature 100°F. Take end grain factor = 1.



- (A) 91 lbs
(B) 117 lbs
(C) 141 lbs
(D) 155 lbs
50. What is the axial stress acting in the wall at mid height due to a 20 kips load acting on a 5 in X 5 in bearing plate on the top? The masonry units are standard hollow 8X8X16 CMU with face shell thickness of 1.25 in and web shell thickness of 1 in. Mortar is not provided at the webs of the units.

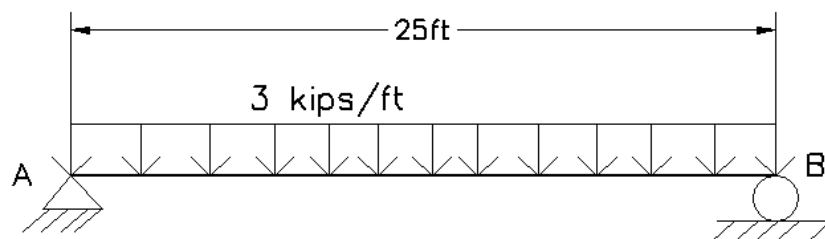


- (A) 133 psi
(B) 157 psi
(C) 195 psi
(D) 250 psi

51. Two rebars shown below are identified as ASTM A615 and ASTM A706. Which of the following is true about the rebars?



- (A) Rebar with marking C denotes carbon steel Rebar
 - (B) Rebar with marking S denotes ASTM A706 low-alloy steel rebar.
 - (C) ASTM A615 rebars are more suitable for seismic reinforcement than A706 rebars because they are stronger than A706 rebars.
 - (D) ASTM A615 rebars are more affordable and less ductile than A706 rebars.
52. After a standard penetration test, which of the following properties of the soil can be correlated with the corrected number of blows 'N' ?
- I) Unconfined compressive strength
 - II) Relative density
 - III) Angle of internal Friction
- (A) I and II
 - (B) II only
 - (C) I, II and III
 - (D) I only
53. A simply supported steel beam has a span of 25 ft and is braced at the midpoint. The beam is subjected to a uniformly distributed live load of 3 kip/ft and bends about the major axis. What is the lightest Gr. 50 section that will satisfy the flexural strength requirements?



- (A) W18X40
- (B) W18X50
- (C) W18X55
- (D) W18X60

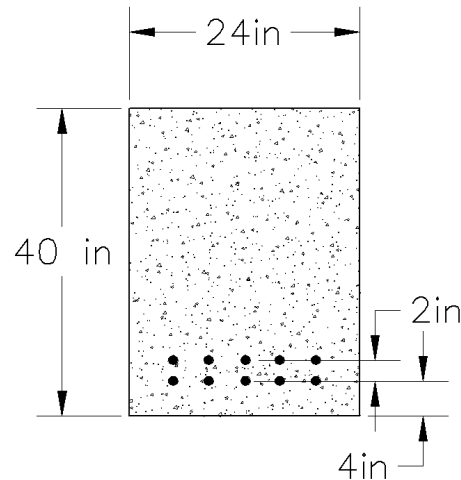
54. Standard wide flange sections are used as horizontal members in a steel construction project to resist transverse loads. Transverse stiffeners are not used in the S beams. The analysis results show that maximum service level shear acting in the beam is 46 kips. What is the most economical section that can be used in the project? All beams have sufficient axial and flexural strengths.



- (A) S6X17.25
 - (B) S8X18.4
 - (C) S8X23
 - (D) S10X25.4
55. A fully grouted reinforced masonry beam is subjected to a shear force of 13 kips. The compressive strength of masonry is 1500 psi. What is the minimum dimension of the masonry beam that satisfies the shear force requirement? Assume $M/Vdv > 1$.
- (A) $b = 7.625$ in , $dv = 16$ in
 - (B) $b = 7.625$ in , $dv = 24$ in
 - (C) $b = 7.625$ in , $dv = 32$ in
 - (D) $b = 7.625$ in , $dv = 40$ in

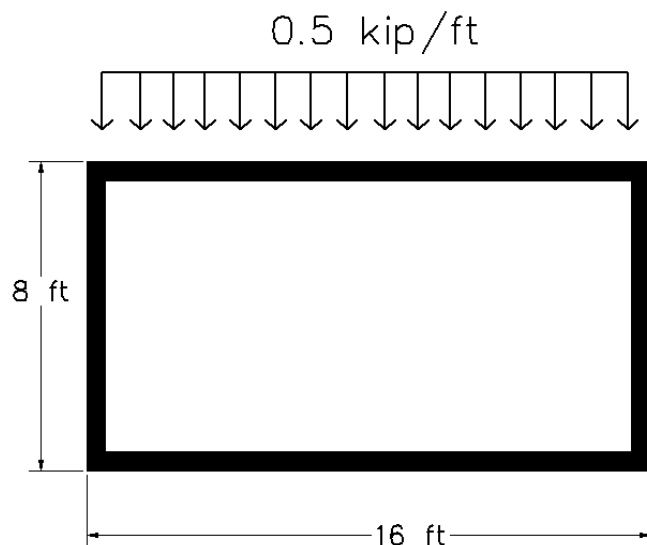
56. A prestressed girder section as shown below only has prestressing strands and no mild steel tensile reinforcement. The compressive stress at the top fiber of the beam due to self weight is 0.75 ksi. The strands are 1/2in diameter low relaxation strands. Area of each strand is 0.153 in². After initial losses, the stress at release per strand is 160 ksi. What is the total top fiber stress at release at the midspan?

- (A) 0 ksi
(B) 0.35 ksi tensile
(C) 0.43 ksi compressive
(D) 0.43 ksi tensile

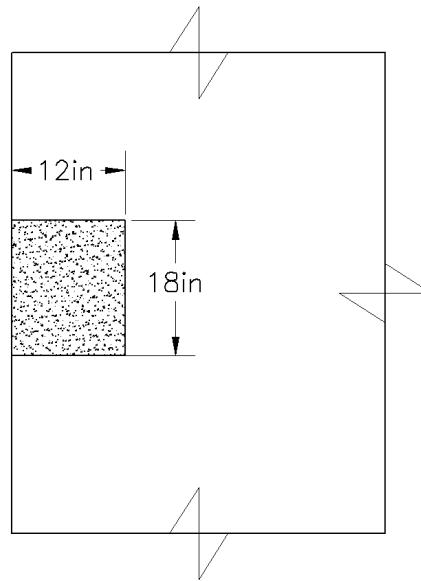


57. The figure below shows a plan of a single story wood frame with Southern pine framing and blocked Structural I, 3/8 inch thick WSP diaphragm. The seismic loading is 0.5 kip/ft. What is the allowable strength of the diaphragm if 8d common nails are used at the diaphragm boundaries with a spacing of 4 inches. Assume nominal width of nailed face at panel edges and boundaries = 2 inches and dry conditions.

- (A) 360 plf
(B) 720 plf
(C) 1060 plf
(D) 1200 plf



58. The plan below shows a column at the edge of a floor slab. The thickness of the slab is 8in and the effective cover is 1.5in. What is the design punching shear capacity of the slab at the location around the edge column? The compressive strength of the concrete slab is 6 ksi.



- (A) 59 kips
(B) 72 kips
(C) 83 kips
(D) 103 kips
59. A pinned-fixed HSS column is subjected to a concentric unfactored axial compression of 15 kips. The height of the column is 13.5 ft and is unbraced in both directions. What is the lightest section that can withstand the applied load?



- (A) HSS 3 X 3 X 3/16
(B) HSS 3 X 3 X 1/4
(C) HSS 3 X 3 X 5/16
(D) HSS 3 X 3 X 3/8

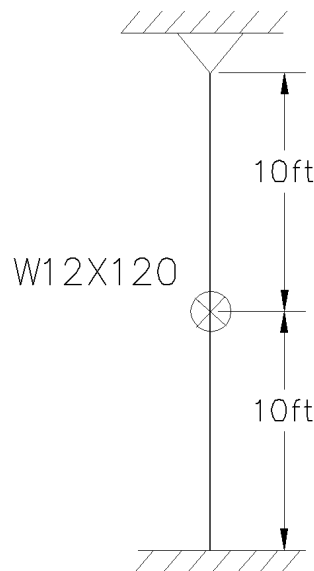
60. A W section column is centered on a 15 in X 15 in base plate. The plate is centered on a 18inX18in concrete pedestal. The maximum concentric axial compression acting on the plate is 1000 kips. What is the minimum strength of concrete pedestal required to bear the load safely using AISC specifications?

- (A) 4.5 ksi
- (B) 7.5 ksi
- (C) 8.5 ksi
- (D) 10.5 ksi

61. The W12X120 column shown below is braced at mid height to prevent buckling against the weaker axis only. What is the governing slenderness ratio KL/r of the column?

$E = 29000$ ksi

$F_y = 50$ ksi

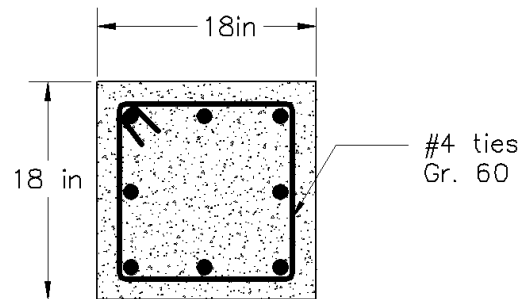


- (A) 34
- (B) 38
- (C) 44
- (D) 48

62. A single story building uses Southern Pine wood framed $3/8$ in thick, blocked Structural -I WSP shear wall. 8d nails with spacing 2 inches are used at the panel edges. What is the allowable seismic shear capacity of the shear wall? Assume dry conditions.

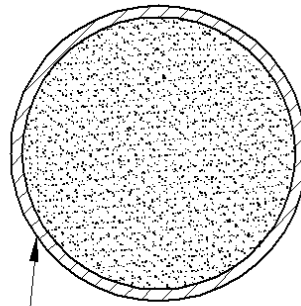
- (A) 610 plf
- (B) 1220 plf
- (C) 1330 plf
- (D) 1500 plf

63. The column shown below is subjected to a factored shear of 75 kips. What is the required tie spacing to satisfy the strength requirement. Assume that the entire shear is taken by the reinforcement only. Effective cover for longitudinal bars = 3 inches.



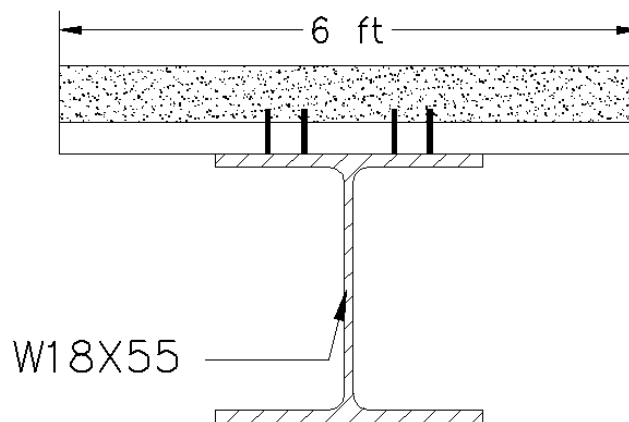
- (A) 3.5 in
(B) 4.75 in
(C) 6 in
(D) 9 in
64. A building frame consists of a combined dual wall and moment frame system to resist lateral loads. What is the primary purpose of the moment frame during a seismic event?
- (A) Transfer lateral loads directly to foundation,
(B) Protect the shear walls from uncertainty in seismic loads,
(C) Resist all the vertical loads,
(D) Enhance lateral stiffness and resistance while the stiffer wall takes the majority of the lateral load.

65. Gr 46 HSS 14.000X0.5 is filled with concrete ($f'_c = 4\text{ksi}$). What is the allowable compressive strength of the composite column section? Elastic critical buckling load of the section is 600 kips. Modulus of elasticity of steel, $E_s = 29000\text{ ksi}$



HSS14.000X0.5

- (A) 263 kips
 (B) 318 kips
 (C) 450 kips
 (D) 526 kips
66. A composite steel beam has four studs in a row as shown below, distributed throughout the span. The beam is subjected to a dead load moment of 150 kip*ft and live load moment of 290 kip*ft. The distance between the top of the steel beam and the concrete compression flange force is 3.5 in. Which of the following is most nearly the total nominal shear carried by the studs due to the applied moments? The capacity of the steel beam and concrete slab is sufficient.

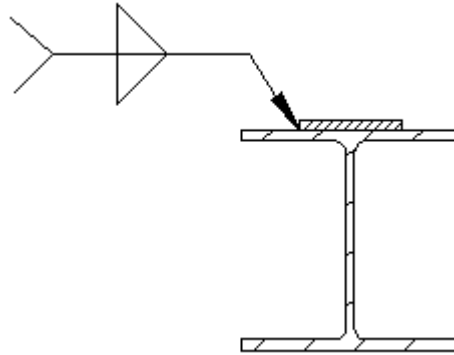


- (A) 335 kips
 (B) 450 kips
 (C) 638 kips
 (D) 750 kips

67. A reinforced concrete beam in a special moment frame has a width of 12 in. and a total depth of 18 in. The beam is designed per ACI 318-14. The negative moment reinforcement at the face of the column consists of four #8 bars. What is the minimum required positive moment reinforcement at the same location to satisfy ACI 318-14 requirements?

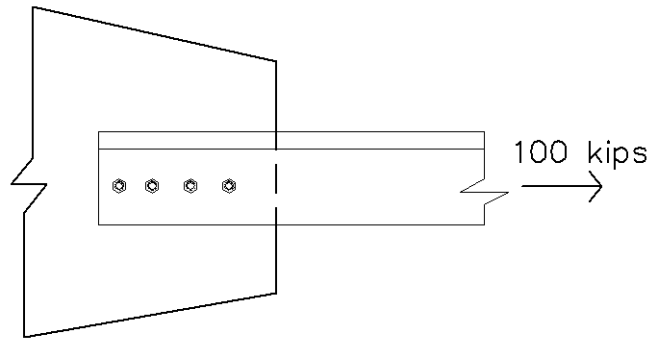
- (A) Six #4 bars.
- (B) Four #5 bars.
- (C) Four #6 bars.
- (D) Four #8 bars.

68. A 0.5 inch thick plate is to be welded on a W section flange as shown below. What is the minimum and maximum size of welds allowable? The flange thickness is greater than the plate thickness.

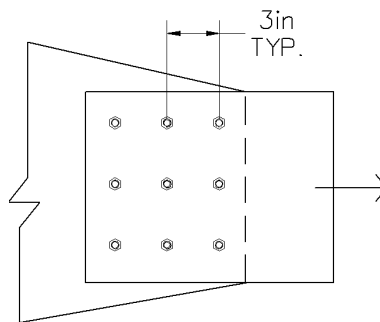


- (A) Min 1/8 in, Max 1/2 in
- (B) Min 5/16 in, Max 1/2 in
- (C) Min 3/16 in, Max 7/16 in
- (D) Min 1/4 in, Max 7/16 in

69. An angle section is connected to a plate with four bolts as shown below. Assuming that the failure of the connection is gross yielding of the section, what is the most economical angle section that satisfies the design requirement? Yield strength of angle is 36 ksi. Loads shown are service loads.



- (A) L 5 X 3 X $\frac{1}{2}$
 (B) L 5 X $3\frac{1}{2}$ X $\frac{1}{2}$
 (C) L 5 X 5 X $\frac{1}{2}$
 (D) L 5 X 5 X $\frac{3}{4}$
70. Two plates are connected as shown below using nine $\frac{7}{8}$ inch diameter A325 bolts with standard holes. Threads are excluded from the shear plane. The thickness of the top plate is $\frac{1}{4}$ in. What is the allowable strength of the connection? Assume plates have sufficient strength for yielding, rupture and block shear and sufficient edge distances. Tensile strength of the plate is 58 ksi.

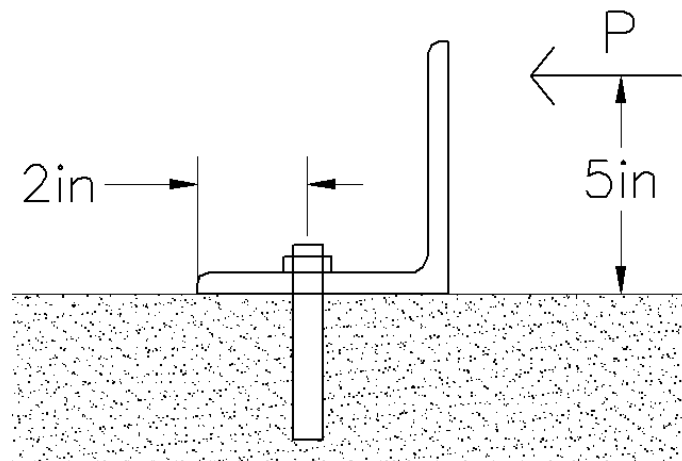


- (A) 137 kips
 (B) 183 kips
 (C) 210 kips
 (D) 255 kips

71. Which of the following statements best describes the function of designing a bolted connection to be slip critical instead of bearing connection?

- (A) Slip critical connections with high pretension and friction allow for increased shear resistance, compared to bearing type connections with same bolts,
- (B) Slip critical connections are used to prevent slip under service loads for stiffness and fatigue requirements.
- (C) Slip critical connections allows for larger bolt hole tolerances during fabrication,
- (D) Slip critical connections are designed to eliminate the need for bolt bearing and tearout checks.

72. An angle section 6X6X1/2 is connected to a concrete slab with a bolt as shown below. The length of the angle section is 4 inches. What is the maximum unfactored load P that can be applied to the angle section as shown below? Assume that the bolts and concrete do not govern the failure and the yield strength of the angle section is 36 ksi.

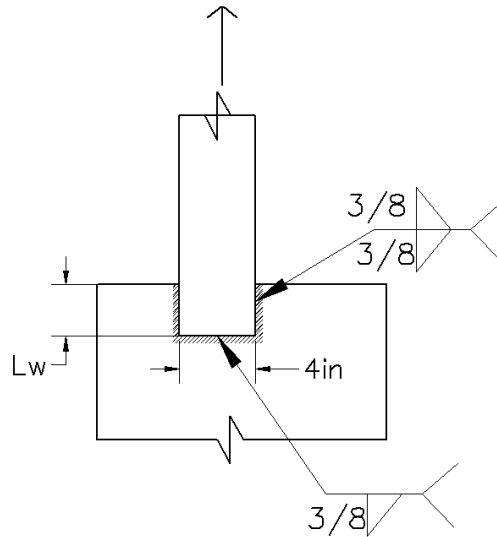


- (A) 0.75 kips
- (B) 1.2 kips
- (C) 1.5 kips
- (D) 1.75 kips

73. A plate is connected to a gusset by welding on two sides as shown below. The service tension load acting concentrically on the welds are 20 kips dead load and 30 kips live load. Assuming that the base metal failure does not govern, what is the required length of weld "Lw" ?

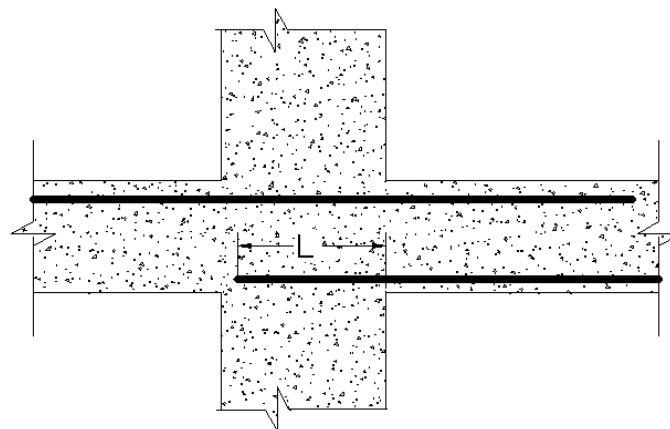
$F_{exx} = 70$ ksi

- (A) 1.5 in
- (B) 2 in
- (C) 2.25 in
- (D) 3 in



74. The figure below shows a continuous beam column joint. The bottom bars are #8 Gr. 60 deformed bars and the effective depth of the beam is 21 inches. Compressive strength of concrete is 4 ksi. There is no positive moment at the face of the column. The point of inflection is 2.5 ft away from the right face of the column. What is the minimum required extension 'L' of bottom bars from the face of the column? The beam is not a part of the primary lateral force resisting system.

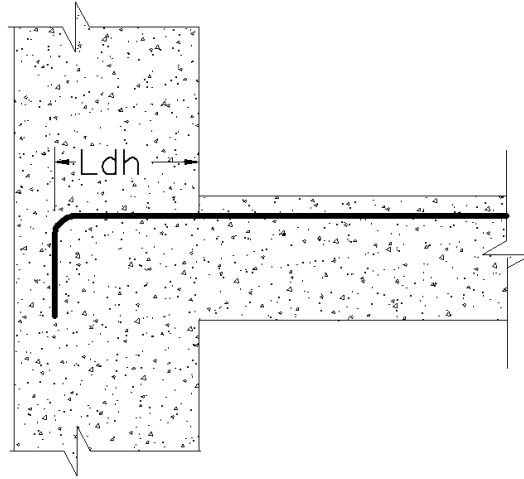
- (A) 0 in
- (B) 6 in
- (C) 12 in
- (D) 21 in



75. The figure below shows a beam column connection for an exterior column. The Gr. 60 #6 bar is hooked into a column that continues to the floors below. What is the minimum required hook development length 'L_{dh}' as shown below?

$f'_c = 4000 \text{ psi}$, $\Psi_e = 1$, $\lambda = 1$

- (A) 6 in
- (B) 10 in
- (C) 12 in
- (D) 18 in



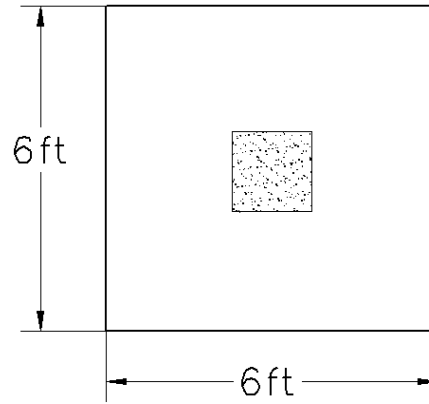
76. A shallow strip footing is bearing on a soil layer with no cohesion. The footing is 2.5 ft wide and located at a depth of 5ft from the ground level. The density of soil is 110 pcf. The water table is measured at a depth 2.5 ft from the ground level. What is the ultimate bearing capacity of the foundation? The angle of internal friction of the soil is 30° .

- (A) 9130 psf
- (B) 12350 psf
- (C) 14700 psf
- (D) 15200 psf

77. Which theoretical assumption is most commonly used when designing shallow foundations for soils that can deform under stress?

- (A) The soil behaves as a rigid body with infinite stiffness.
- (B) The soil can be modeled as a series of independent linear springs.
- (C) The soil has no shear strength and relies solely on cohesion.
- (D) The footing is infinitely flexible and does not distribute loads.

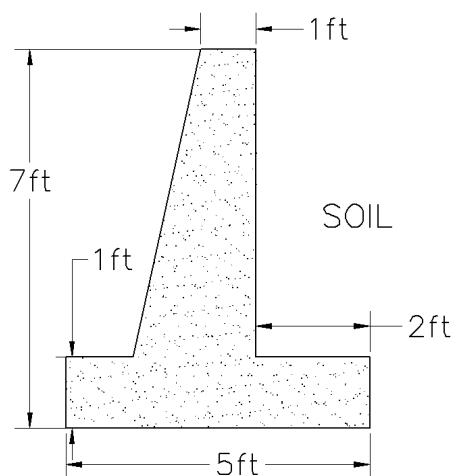
78. The plan of a square RCC footing is shown below. The critical factored shear acting on the footing is 150 kips. The effective cover of the bottom flexural bars is 3 in and there are no shear reinforcements. What is the required depth of footing for one way shear strength?
 $f'_c = 4 \text{ ksi}$ $\lambda = 1$



- (A) 18 in
 (B) 22 in
 (C) 25 in
 (D) 28 in

79. What is negative skin friction in piles and when does it occur in piles?
 (A) It is the downward drag on the piles due to consolidation of surrounding soil,
 (B) It is the additional uplift force in the piles acting in the negative direction of applied load.
 (C) It is the reduction in the frictional bearing capacity of the piles due to pile driving methods,
 (D) It is the skin friction acting in the negative direction during extraction of piles.

80. A cantilevered wall is retaining soil on one side only as shown below. The height of the backfill is 7 ft. What is the maximum factored flexural demand on the stem of the wall?
 Density of soil = 110 pcf Active earth pressure coefficient, $K_a = 0.33$



- (A) 1100 lb*ft/ft
 (B) 1306 lb*ft/ft
 (C) 1893 lb*ft/ft
 (D) 2091 lb*ft/ft

Solutions

[Click here for Problems](#)

1. Problem

Correct Answer: (C)

Explanation:

Per ASCE 7-16, section 3.1.4 'The weight of all landscaping and hardscaping materials shall be considered as dead load. The weight shall be computed considering both fully saturated soil and drainage layer materials and fully dry soil and drainage layer materials to determine the most severe load effects on the structure.'

Hence, option C that says the average moisture content of soil should be taken, is against the code and is not true.

2. Problem

Correct Answer: (C) 4067 kips

Explanation:

{Ref IBC 2018 page 368, Table 1607.1 for provisions for staircase live loads}

Uniformly distributed staircase load to be considered for school buildings=100 psf

Calculate total area of horizontal projection of stairs and landing.

No of treads = 5

Depth of treads = 10 inches

Width of stairs and landings = 4ft

Area of horizontal projection of treads = $5 \times 10 \text{ inches} \times 4 \text{ ft} = 16.67 \text{ ft}^2$

Area of two landings = $2 \times 3 \text{ ft} \times 4 \text{ ft} = 24 \text{ ft}^2$

Total area = $24 \text{ ft}^2 + 16.67 \text{ ft}^2 = 40.67 \text{ ft}^2$

Total load = $100 \text{ psf} \times 40.67 \text{ ft}^2$

=4067 lb

3. Problem

Correct Answer: (B) 3.125 kips/ft

Explanation:

{Ref ASCE 7-16 page 15, Table 4.3-1 for provisions for gymnasium live loads}

Uniformly distributed load to be considered for gymnasiums $L_0=100$ psf

Tributary Area, $A_T=1250$ ft²

Per ASCE 7-16 page 17, section 4.7.5, for assembly usage, live loads cannot be reduced.

Hence, total live load acting on the beam = $A_T * L_0$

= $1250 \text{ ft}^2 * 100 \text{ psf}$

= 125000 lb

= 125 kips

Uniformly distributed load

= 125 kip/Length of beam

= 125 kip/40ft

= 3.125 kips/ft

4. Problem

Correct Answer: (C) 4ft

Explanation:

Construction load = 50 psf

Density of slab = 150 pcf

Thickness of slab = 6 inch = 0.5 ft

Dead load = 150 pcf * 0.5 ft = 75 psf

Total load during concrete placement = construction load + self-weight of slab

= 75 psf + 50 psf

= 125 psf

Allowable carrying capacity of floor props = 2 kips = 2000 lb

Area that can be safely supported = prop capacity / loading intensity

= $2000 \text{ lb} / (125 \text{ lb} / \text{ft}^2)$

= 16 ft²

Maximum permitted spacing in each direction = $\sqrt{16} \text{ ft}$

S = 4ft

5. Problem

Correct Answer: (C) 15.24 lb/ft²

Explanation:

Given Basic Wind Speed = 90 mph

Velocity pressure is given by ASCE 7-16 equation 26.10-1

$$q_z = 0.00256 K_z K_{zt} K_d K_e V^2$$

Since topographic effects can be neglected, $K_{zt} = 1$

From Table 26.9-1,

For ground elevation above 2000 ft, elevation factor $K_e = 0.93$

From Table 26.10-1,

For 80ft, Exposure B, Velocity pressure coefficient $K_z = 0.93$

From Table 26.6-1, Wind directionality factor, $K_d = 0.85$ for Buildings MWFRS

$$q_z = 0.00256 * 0.93 * 1 * 0.85 * 0.93 * 90^2$$

$$q_z = 0.00256 * 0.93 * 1 * 0.85 * 0.93 * 90^2$$

$$q_z = 15.24 \frac{lb}{ft^2}$$

6. Problem

Correct Answer: (D) Seismic Design category D

Explanation:

Seismic design category is determined following procedures of ASCE 7-16 Chapter 11

Mapped short period spectral acceleration $S_s = 1g$

Mapped 1s period spectral acceleration $S_1 = 0.4g$

Per ASCE 7-16 page 204, table 20.3-1, the site with stiff soil is classified as class D.

From Page 84, table 11.4-1,

For class D and $S_s = 1$, $F_a = 1.1$

From Page 84, table 11.4-2,

For class D and $S_1 = 0.4$, $F_v = 1.9$

Calculate MCE_R parameters adjusted for site class from Section 11.4.4

$$S_{MS} = F_a * S_s$$

$$= 1.1 * 1g = 1.1g$$

$$S_{M1} = F_v * S_1$$

$$= 1.9 * 0.4g = 0.76g$$

Calculate Design spectral acceleration parameters per Section 11.4.5 page 84

$$S_{DS} = (2/3) * S_{MS} = 0.667 * 1.1g = 0.73g$$

$$S_{D1} = (2/3) * S_{M1} = 0.667 * 0.76g = 0.5g$$

From page 85 Table 11.6-1 For Risk category IV, for $S_{DS} > 0.5$, seismic design category is D

From page 85 Table 11.6-2 For Risk category IV, for $S_{D1} > 0.2$, seismic design category is D

Hence correct answer is seismic design category D

7. Problem

Correct Answer: (C) 0.55 seconds

Explanation:

Per ASCE 7-16, page 102 section 12.8.2.1, the approximate time period of framed structures can be calculated using Equation 12.8-7

Structural height of building, $h_n = 50\text{ft}$

From Table 12.8-2

$C_t = 0.016$ for concrete moment resisting frames

$x = 0.9$

$$T_a = C_t * h_n^x$$

$$T_a = 0.016 * 50^{0.9} = 0.54 \text{ seconds} \approx \mathbf{0.55 \text{ seconds}}$$

8. Problem

Correct Answer: (C) 31.4 kips

Explanation:

Reaction due to uniform lane load is constant regardless of the position of the concentrated load

$$R_1 = (0.64 \text{ kip/ft} * 20 \text{ ft}) / 2$$

$$= 12.8 \text{ kip} / 2$$

$$= 6.4 \text{ kips}$$

Reaction due to moving concentrated load will be maximum when the concentrated load is directly on top of the supports.

It will be equal to the concentrated load itself

$$R_2 = 25 \text{ kips}$$

$$\text{Total reaction} = R_1 + R_2 = 25 \text{ kips} + 6.4 \text{ kips} = 31.4 \text{ kips}$$

Thus maximum possible reaction on either supports = **31.4 kips**

9. Problem

Correct Answer: (C)

Explanation:

From ASCE 7-16 page 62, section 7.10, all other conditions are valid except statement C.

Slope of roof = 20 degrees

$$W/50 = 20/50 = 0.4$$

Since $20 > 0.4$ rain surcharge does not apply and should be 0 psf.

10. Problem

Correct Answer: (D) **21.6 kip*ft**

Explanation:

Per ASCE 7-16 section 4.5.3, page 16

Load to be considered for passenger vehicle barrier

$$P = 6 \text{ kips}$$

This load is assumed to act at a height of 1ft 6in to 2ft 3in from floor level

For maximum moment, the load is applied at 2ft 3in above floor in the cantilevered barrier

$$\text{Moment} = P * 2\text{ft}3\text{in} = 6\text{kip} * 2\text{ft}3\text{in} = 13.5 \text{ kip*ft}$$

Load factor for vehicle collision loads is the same as for live loads.

$$\text{Thus, factored moment} = 1.6 * 13.5 \text{ kip*ft} = \mathbf{21.6 \text{ kip*ft}}$$

11. Problem

Correct Answer: (D) **2.4 kips/ft**

Explanation:

Active earth pressure coefficient, $K_a = 0.33$

Dry density of soil, $\gamma = 110 \text{ pcf}$

Submerged density of soil, $\gamma' = 65 \text{ pcf}$

Density of water, $\gamma_w = 62.4 \text{ pcf}$

Height of water table = 5ft

Pressure intensity due to dry soil upto 5ft depth

$$= K_a * \gamma * 5\text{ft}$$

$$= 0.33 * 110\text{pcf} * 5\text{ft} = 181.5 \text{ psf}$$

Pressure intensity due to submerged soil below 5ft

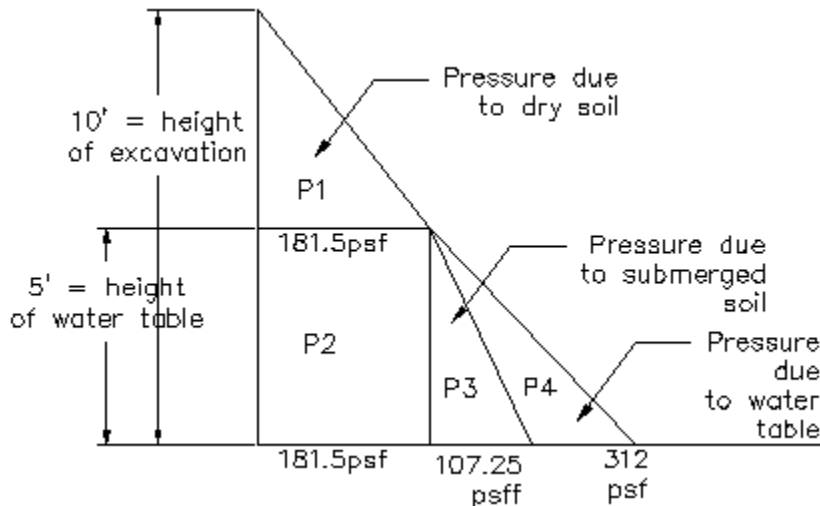
$$= K_a * \gamma' * (10\text{ft} - 5\text{ft})$$

$$=0.33*65\text{pcf}*5\text{ft}=107.25\text{psf}$$

Pressure intensity due to water table below 5ft

$$= \gamma_2*(10\text{ft}-5\text{ft})$$

$$=62.4\text{pcf}*5\text{ft}=312\text{psf}$$



$$P1 = 0.5*5\text{ft}*181.5 \text{ psf} = 453.75 \text{ plf}$$

$$P2 = 5\text{ft}*181.5\text{psf} = 907.5 \text{ plf}$$

$$P3 = 0.5*5\text{ft}*107.25\text{psf} = 268.125 \text{ plf}$$

$$P4 = 0.5*5\text{ft}*312\text{psf} = 780\text{plf}$$

Total load is the sum of all pressure areas

$$P1+P2+P3+P4 = (435.75+907.5+268.125+780)\text{plf} = 2391.375 \text{ lb/ft} \approx \mathbf{2.4 \text{ kips/ft}}$$

12. Problem

Correct Answer: (C)

Explanation:

All options are correct except option C.

Active earth pressure can be positive or negative for non granular soil with cohesion and develops tension cracks.

13. Problem

Correct Answer: (B) Diagonals in Compression, vertical members in tension, upper chord in compression and bottom chord in tension,

Explanation:

All options are incorrect except option B.

This is a typical Howe bridge truss. A passing car exerts gravity loads on the truss. When there is no lateral load, the diagonals are in compression, the vertical members are in tension, the upper chord is in compression and the bottom chord is in tension.

14. Problem

Correct Answer: (B) 2328 kip*ft

Explanation:

From AASHTO LRFD 2017 page 3-21, Table 3.6.1.1.2-1 multiple presence factor for 1 lane Bridge, $m = 1.2$

From page 3-31, Table 3.6.2.1-1, impact factor for truck loads $IM = 33\%$

From page 3-16, Table 3.4.1-2, for dead loads, max factor $= 1.25$

From page 3-15, Table 3.4.1-1, for live loads, max factor $= 1.75$

Dead load moment $= 800 \text{ kip*ft}$

Live load moment including multiple presence factor and impact factor

$= 1.2 * (250 \text{ kip*ft} * 1.33 + 300 \text{ kip*ft}) = 759 \text{ kip*ft}$ (impact for truck only)

Thus strength I combination $= 1.25 * 800 \text{ kip*ft} + 1.75 * 759 \text{ kip*ft} = 2328 \text{ kip*ft}$

15. Problem

Correct Answer: (B)

Explanation:

For the same cross sectional area, the hollow tube will have the highest polar moment of inertia and hence will have the best torsional rigidity out of all the shapes. Thus it is best suited and the correct answer is option B.

16. Problem

Correct Answer: (A)

Explanation:

When a lateral load acts in the direction shown, the left column will be in tension and the right column will be in compression. The center column will have very little axial force if any. Hence option B is ruled out. The beam will be in axial compression. As the axial load is transferred in the beam from the left bay to the right bay, a part of the axial load will be transferred to the center column as shear loads. Hence the axial compression in the right beam will be lesser than the left beam. Thus the correct answer is option A.

17. Problem

Correct Answer: (C)

Explanation:

The vertical member CG is a zero force member unless a vertical load acts directly on top of the member at node C. Here, the load is travelling along the bottom chord between nodes A,H,G,F and E. For all these cases, the force in member CG will be zero. Hence the correct answer is Option C that shows zero force throughout.

18. Problem

Correct Answer: (C) 0.295 in

Explanation: Ref NCEES PE handbook page 29, section 1.6.3 for thermal deformations.

$$A = 20 \text{ in}^2$$

$$L = 30 \text{ ft}$$

$$\alpha = 0.000007 / ^\circ\text{F}$$

$$P = 200 \text{ kips}$$

$$\Delta T = 100 - 32 = 68^\circ\text{F}$$

Ref to NCEES PE handbook, page 28, section 1.6.2.6, for linearly elastic material,

$$E = \frac{\text{Stress}}{\text{Strain}}$$

Change in length due to load 'P'

$$\Delta L1 = \frac{PL}{AE} = \frac{200 * 30 \text{ ft} * 12 (\text{in/ft})}{20 \text{ in}^2 * 29000 \text{ ksi}} = 0.124 \text{ in}$$

Change in length due to load 'ΔT'

$$\Delta L2 = \alpha * L * \Delta T = 0.000007 / ^\circ\text{F} * 30 \text{ ft} * 12 (\text{in/ft}) * 68^\circ\text{F} = 0.171 \text{ in}$$

Total change in length = $\Delta L_1 + \Delta L_2 = 0.124 + 0.171 = \mathbf{0.295 \text{ in}}$

Correct answer is option C.

19. Problem

Correct Answer: (B) 92.3 kips

Explanation: Ref NCEES PE handbook page 29, section 1.6.3, for thermal deformations.

$$A = 15 \text{ in}^2$$

$$L = 30 \text{ ft}$$

$$\alpha = 0.000007 / ^\circ\text{F}$$

$$\Delta T = 100 - 30 = 70^\circ\text{F}$$

Change in length due to change in temperature ' ΔT '

$$\Delta L = \alpha * L * \Delta T = 0.000007 / ^\circ\text{F} * 30 \text{ ft} * 12 (\text{in/ft}) * 70^\circ\text{F} = 0.1764 \text{ in}$$

There is clearance of only 0.1 inch for expansion.

The restriction in expansion due to insufficient clearance induces thermal compressive stresses

$$\Delta L_r = 0.1764 \text{ in} - 0.1 \text{ in} = 0.0764 \text{ in}$$

For linearly elastic material,

$$E = \frac{\text{Stress}}{\text{Strain}}$$

Change in length due to load 'P'

$$E = \frac{P/A}{\Delta L_r/L}$$

$$P = E * A * \Delta L_r / L = 29000 \text{ ksi} * 15 \text{ in}^2 * 0.0764 \text{ in} / \{30 \text{ ft} * 12 (\text{in/ft})\} = \mathbf{92.3 \text{ kips}}$$

Correct answer is option B.

20. Problem

Correct Answer: (A) 112 kips (compression)

Explanation:

Calculate reaction at joint D

Sum of Moments at joint A = 0

$$R_d \cdot (4+8+3) = 80 \cdot 4 + 40 \cdot 6 + 60 \cdot (4+8)$$

$$R_d = 1280/15 = 85.33 \text{ kips}$$

Angle of inclination of member CD

$$\Theta = \tan^{-1} (3.5/3) = 49.4^\circ$$

Analyze joint D

Vertical component of F_{cd}

$$F_{y_{cd}} = F_{cd} \cdot \sin(49.4)$$

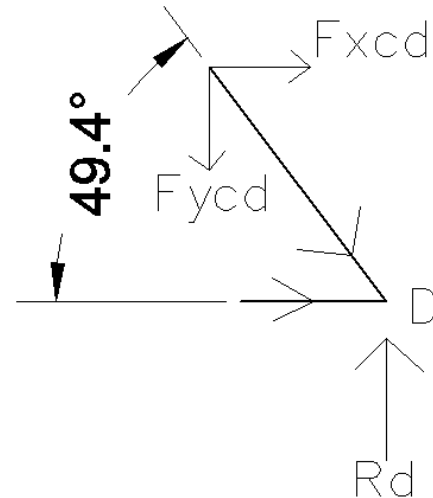
Taking sum of vertical forces at joint D = 0

$$F_{y_{cd}} = R_d$$

$$F_{cd} \cdot \sin(49.4) = 85.33 \text{ kips}$$

$F_{cd} \approx \mathbf{112 \text{ kips (compression)}}$

Hence the correct answer is option A.



21. Problem

Correct Answer: (B) BH = 0 DF = 0

Explanation:

When we analyze joint H, there is no external load. BH is the only vertical member.

So when we take sum of forces in vertical direction = 0,

$$\text{Force in BH} = 0$$

When we analyze joint F, there is no external load. DF is the only vertical member.

So when we take sum of forces in vertical direction = 0,

$$\text{Force in DF} = 0$$

Hence both BH and DF are zero force members and the correct answer is option B. .

22. Problem

Correct Answer: (D) 2943 psi

Explanation:

Ref to NCEES PE handbook page 20,21 for centroid, moment of inertia and page 32 for shear stress distribution in beams

Calculate location of neutral axis

Area of flange, $A_f = 4 \text{ in} \times 0.5 \text{ in} = 2 \text{ in}^2$

Moment of inertia of flange, $I_f = \{4 \text{ in} \times (0.5 \text{ in})^3\}/12 = 0.042 \text{ in}^4$

Location of centroid of flange from base, $y_f = 2 \text{ in} + 0.25 \text{ in} = 2.25 \text{ in}$

Area of web, $A_w = 2 \text{ in} \times 0.5 \text{ in} = 1 \text{ in}^2$

Moment of inertia of web, $I_w = \{0.5 \text{ in} \times (2 \text{ in})^3\}/12 = 0.333 \text{ in}^4$

Location of centroid of web from base, $y_w = 2 \text{ in}/2 = 1 \text{ in}$

$y' = (2 \text{ in}^2 \times 2.25 \text{ in} + 1 \text{ in}^2 \times 1 \text{ in}) / (2 \text{ in}^2 + 1 \text{ in}^2) = 1.833 \text{ in}$

Calculate total moment of inertia about centroid

$$\begin{aligned} I_{xx} &= I_f + A_f(y_f - y')^2 + I_w + A_w(y' - y_w)^2 \\ &= 0.042 + 2(2.25 - 1.833)^2 + 0.333 + 1(1.833 - 1)^2 \\ &= 1.417 \text{ in}^4 \end{aligned}$$

Calculate first moment of area above the junction of web and flange, about the centroid,

$$Q = 4 \text{ in} \times 0.5 \text{ in} \times (2.25 \text{ in} - 1.833 \text{ in}) = 0.834 \text{ in}^3$$

Thickness of web, $b = 0.5 \text{ in}$

Shear force, $V = 2.5 \text{ kips} = 2500 \text{ lbs}$

Shear stress,

$$\begin{aligned} \tau &= (VQ)/(Ib) \\ &= (2500 \text{ lbs} \times 0.834 \text{ in}^3) / (1.417 \text{ in}^4 \times 0.5 \text{ in}) \\ &= \mathbf{2943 \text{ psi}} \end{aligned}$$

Thus correct option is option D

23. Problem

Correct Answer: (D) 5 kips/in

Explanation:

Here, the load is concentric to the centroid of the weld group.

So, there is no torsion or moments causing additional shear on the welds.

Total length of the welds (L)

$$= 6 \text{ in} + 6 \text{ in} + 4 \text{ in}$$

$$= 16 \text{ in}$$

$$\text{Load} = 80 \text{ kips}$$

Shear on welds

$$= 80 \text{ kips} / 16 \text{ in}$$

$$= \mathbf{5 \text{ kips/in}}$$

Correct answer is Option D.

24. Problem

Correct Answer: (C) -11 kip*ft

Explanation: Ref NCEES PE handbook page 32.

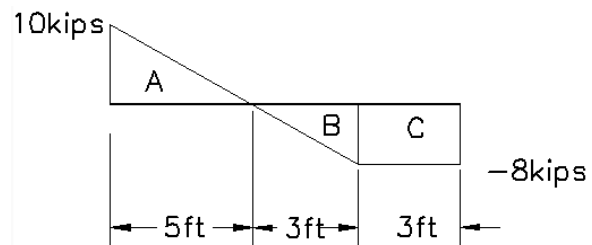
The difference in bending moment between any two points is given by the area of shear force diagram between the two points. Since the beam is simply supported, the moment at the left end is zero. So the value of the bending moment at 11 ft is given by the sum of areas of triangles A, B and rectangle C.

Total Area

$$= 0.5 * 10 \text{ kips} * 5 \text{ ft} - 0.5 * 8 \text{ kips} * 3 \text{ ft} - 8 \text{ kips} * 3 \text{ ft}$$

$$= \mathbf{-11 \text{ kip*ft}}$$

Correct Answer is Option C.



25. Problem

Correct Answer: (C) + 6.4 kip*ft , -8.4 kip*ft

Explanation:

Ref to ACI 318-14 page 70 for simplified method of analysis for continuous beams.

Uniform loading, $w = 0.4$ kips/ft

Calculate maximum positive moment

For longest end span

$$M_1 = w \cdot 15\text{ft} \cdot 15\text{ft} / 14 = 6.43 \text{ kip*ft}$$

For longest interior span,

$$M_2 = w \cdot 14\text{ft} \cdot 14\text{ft} / 16 = 4.9 \text{ kip*ft}$$

Calculate maximum negative moment

Based on the dimensions given, critical section is at exterior face of first interior support (from the right),

Average of end spans 14ft and 15ft = 14.5ft

$$M_3 = w \cdot 14.5\text{ft} \cdot 14.5\text{ft} / 10 = 8.41 \text{ kip*ft}$$

Hence max positive moment is **6.4 kip*ft** and max negative moment is **- 8.4 kip*ft**.

Correct answer is option C

26. Problem

Correct Answer: (C) 35.37 kip*ft

Explanation:

Ref NCEES PE handbook page 241-244 for FEMs.

Due to 10 kip point load at 7ft from joint C,

$$FEM_{CB} = \frac{Pa^2b}{L^2} = 10 \text{ kip} \cdot (3 \text{ ft} \cdot 3\text{ft}) \cdot 7\text{ft} / (10\text{ft} \cdot 10\text{ft}) = 6.3 \text{ kip*ft}$$

Due to distributed load 5 kip/ft

$$FEM_{CD} = \frac{wL^2}{12} = -5 \text{ kip/ft} \cdot (10 \text{ ft} \cdot 10\text{ft}) / 12 = -41.67 \text{ kip*ft}$$

Thus unbalanced moment = 41.67 kip*ft - 6.3 kip*ft = **35.37 kip*ft**

Correct answer is option C.

27. Problem

Correct Answer: (C) 8.2 ksi

Explanation:

Area of the section (A) = 8 in * 16 in = 128 in²

Section modulus about strong axis, $S_x = \{8 \text{ in} * (16 \text{ in})^2\} / 6 = 341.33 \text{ in}^3$

Section modulus about weak axis $S_y = \{16 \text{ in} * (8 \text{ in})^2\} / 6 = 170.67 \text{ in}^3$

P = 150 kip

M_x = 100 kip*ft

M_y = 50 kip*ft

Max normal stress = $P / A + M_x / S_x + M_y / S_y$

= $150 \text{ kip} / 128 \text{ in}^2 + 100 \text{ kip*ft} * 12(\text{in/ft}) / 341.33 \text{ in}^3 + 50 \text{ kip*ft} * 12(\text{in/ft}) / 170.67 \text{ in}^3$

= **8.2 ksi**

Correct answer is option C.

28. Problem

Correct Answer: (D)

Explanation:

Based on the graph of failure envelope, the section fails if the sum of utilization is equal to or exceeds 100%.

For option D, sum of utilization = 60% + 50% = 110%. Hence the section fails and option D is False. Thus, it is the correct answer

29. Problem

Correct Answer: (B) 0.15 in

Explanation:

Ref to AISC steel manual page 1-29 for properties of W10X45.

$I = 53.4 \text{ in}^4$ (about weaker axis)

Ref to NCEES PE handbook page 255 for maximum deflection.

Loading, $w = 2 \text{ kips/ft} = 0.1667 \text{ kips/in}$

Span, $l = 10 \text{ ft} = 120 \text{ in}$

Maximum deflection is given by

$\Delta_{\text{max}} = 0.0065 w l^4 / EI$

= $0.0065 * 0.1667 \text{ kips/in} * (120 \text{ in})^4 / (29000 \text{ ksi} * 53.4 \text{ in}^4)$

= $0.145 \text{ in} \approx \mathbf{0.15 \text{ in}}$

Correct answer is option B.

30. Problem

Correct Answer: (D) 2.12 in

Explanation:

Ref to NCEES PE handbook page 242 for deflections due to uniform loadings.

$$I = 84 \text{ in}^4$$

$$E = 29000 \text{ ksi}$$

$$\text{span, } l = 15 \text{ ft} = 180 \text{ in}$$

$$\text{Uniform loading, } w = 12 \text{ kips/ft} = 1 \text{ kip/in}$$

$$\text{Point load, } P = 80 \text{ kips}$$

Maximum deflection by uniformly distributed load is given by

$$\Delta_1 = w l^4 / (384EI)$$

$$= 1 \text{ kips/in} * (180 \text{ in})^4 / (384 * 29000 \text{ ksi} * 84 \text{ in}^4) = 1.12 \text{ in}$$

Maximum deflection by point load is given by

$$\Delta_1 = P l^3 / (192EI)$$

$$= 80 \text{ kips} * (180 \text{ in})^3 / (192 * 29000 \text{ ksi} * 84 \text{ in}^4) = 1 \text{ in}$$

Both max deflections occur at mid span.

Hence max deflection

$$\Delta_{\text{max}} = \Delta_1 + \Delta_2$$

$$= 1.12 \text{ in} + 1 \text{ in} = \mathbf{2.12 \text{ in}}$$

Correct answer is option D.

31. Problem

Correct Answer: (A) 184/3EI

Explanation:

Ref NCEES PE handbook page 233 for frame deflection by unit load method,

$$\Delta = \sum \int m_i * (M_i / EI) dx$$

m_i is the moment equation in member i due to unit load

M_i is the moment equation in member i due to load applied to frame

For this case, since unit load is acting in the frame, $m_i = M_i$

For member BC, moment equation

$$m_i = 1 \text{ kip} * x = x$$

For member AB,

$$m_i = 1 \text{ kip} * 4 \text{ ft} = 4 \text{ kip} \cdot \text{ft}$$

Moment of inertia of BC = I, limits = 0ft to 4ft

Moment of inertia of AB = 2I, limits = 0ft to 5ft

Then the integral becomes,

$$\Delta = \int_0^4 x^3 dx / (EI) + \int_0^5 4^3 dx / (2EI)$$

$$\Delta = \{(4^3/3-0)/(EI)\} + \{16^3(5-0)/(2EI)\}$$

$$\Delta = 64/3EI + 40EI$$

$$\Delta = 184/3EI$$

Correct answer is option A.

32. Problem

Correct Answer: (A)

Explanation:

Ref NCEES PE handbook page 242 for beam deflections, $\Delta = PL^3/192EI$

When depth of beam is doubled, the new moment of inertia is

$$I_{\text{new}} = b(2h)^3/12 = 8(bh^3/12) = 8 * I_{\text{original}}$$

$$\text{Length, } L_{\text{new}} = 2 L_{\text{original}}$$

Deflection is directly proportional to cube of Length 'L' and inversely proportional to 'I'

$$\Delta_{\text{new}} / \Delta_{\text{original}} = (L_{\text{new}} / L_{\text{original}})^3 * I_{\text{original}} / I_{\text{new}}$$

$$\Delta_{\text{new}} / \Delta_{\text{original}} = (2 L_{\text{original}} / L_{\text{original}})^3 * I_{\text{original}} / (8 * I_{\text{original}})$$

$$\Delta_{\text{new}} / \Delta_{\text{original}} = (8) * (1/8) = 1.$$

Thus deflections will be equal if depth is doubled and the correct answer is option A.

33. Problem

Correct Answer: (A)

Explanation:

Inelastic buckling occurs when the material stress has exceeded its elastic limits and is always greater than the elastic buckling strength. Hence option A is false and is the correct answer. All other options are true.

34. Problem

Correct Answer: (C) Infinite

Explanation:

A constant amplitude fatigue limit is the value of stress range under which the body can theoretically withstand an infinite number of loading cycles. Hence, the correct answer for this question is “Infinite”.

35. Problem

Correct Answer: (A)

Explanation:

Option A is incorrect because although the owner, engineer and building officials from the authority have the right to approve/disapprove the party serving as special inspectors, the contractor has no right to choose who serves as the inspector. All other options are true and not the answer.

36. Problem

Correct Answer: (D)

Explanation:

The submittal document content and all associated procedures are project specific and governed by the contract document and specifications for the associated project. Hence the correct answer is Option D ‘Contract documents and specifications’.

37. Problem

Correct Answer: (B) 6ft X6ft

Explanation:

Horizontal component of cable force, $H = 5 \text{ kips} * \cos (30 \text{ deg}) = 4.33 \text{ kips}$

Vertical component of cable force pulling upward, $V = 5 \text{ kips} * \sin (30 \text{ deg}) = 2.5 \text{ kips}$

Coefficient of friction, $\mu = 0.3$

Let 'W' be the weight of the block

Normal reaction * coefficient of friction = Factor of safety * Horizontal force

$$(W - V) * \mu = 1.5 * H$$

$$W = 1.5 * 4.33 \text{ kips} / 0.3 + 2.5 \text{ kips} = 24.15 \text{ kips}$$

6 ft height of block is constant, Try 6 ft X 5 ft X 5 ft block

Weight = $0.15 \text{ kips/ft}^3 * 6 \text{ ft} * 5 \text{ ft} * 5 \text{ ft} = 22.5 \text{ kips}$, Little less than required.

Try 6 ft X 6 ft X 6 ft block

Weight = $0.15 \text{ kips/ft}^3 * 6 \text{ ft} * 6 \text{ ft} * 6 \text{ ft} = 32.4 \text{ kips}$, Sufficient

Check tipping of block

Overtipping moment = $H * 6 \text{ ft} = 4.33 \text{ kip} * 6 \text{ ft} = 25.98 \text{ kip*ft}$

Restoring moment = $(W - V) * 6 \text{ ft} / 2 = (32.4 \text{ kips} - 2.5 \text{ kips}) * 3 \text{ ft} = 89.7 \text{ kip*ft}$

Thus, we have sufficient factor of safety against overturning.

Hence, 6ft X 6ft X 6ft block will satisfy the requirement and option B is correct.

38. Problem

Correct Answer: (B) 20 kips

Explanation:

Maximum load transferred to 5th floor slab = 50 kips - 35 kips

= 15 kips

Maximum load transferred to 4th floor slab = 35 kips - 15 kips

= 20 kips

Maximum load transferred to 3rd floor slab = 15 kips - 5 kips

= 10 kips

Maximum load transferred to 2nd floor slab = 5 kips - 0

= 5 kips

Thus maximum load is transferred to 4th floor slab = **20 kips**, Option B is the answer.

39. Problem

Correct Answer: (C) 3 kip*ft

Explanation:

Per OSHA, CFR Title 29, Part 1926 ,Subpart L, section 1926.451(a)(1)

The scaffold component shall be capable of supporting its own weight and at least 4 times the maximum intended load.

Maximum intended load on each beam = $20 \text{ psf} \times 5\text{ft}/2 = 50 \text{ plf}$

Design load including self weight of the beam

= $40 \text{ plf} + 4 \times 50\text{plf} = 0.24 \text{ kip/ft}$

Design moment for simply supported beam (Ref NCEES PE handbook page 236 Case 1)

= $wl^2/8 = 0.24 \text{ kip/ft} \times 10 \text{ ft} \times 10 \text{ ft} / 8$

= **3 kip*ft**, Correct answer is option C.

40. Problem

Correct Answer: (B) 4

Explanation:

Per OSHA, CFR Title 29, Part 1926 ,Subpart R, 1926.758(c)

Rigid frames shall have 50 percent of their bolts installed and tightened on both sides of the web adjacent to each flange before the hoisting equipment is released.

Number of bolts on each side of web = 8

Minimum number of bolts to be tightened = $50\% \times 8 = 4$

Correct answer is option B.

41. Problem

Correct Answer: (C) CH

Explanation:

Ref to NCEES PE handbook page 110, section 3.7.2 for USCS classification.

% of particles passing through No. 200 (0.075 mm sieve)

= $(1-0.31) \times 100\%$

= 69%

Hence the inorganic sample is silt and clay.

Liquid Limit = 51 i.e. greater than 50, so it is either CH or MH

From page 111,

$PI = LL - PL$

$$= 51 - 19 = 32$$

LL = 51 and PI = 32 plots above the A line from the plasticity chart on page 111.

Hence from table in page 110, it is classified as **CH**.

Correct answer is Option C.

42. Problem

Correct Answer: (C) 75%

Explanation:

Ref to NCEES PE handbook page 135

Water content w = weight of water/weight of solids

$$w = W_w/W_s$$

$$0.19 = W_w/W_s$$

$$W_w = 0.19W_s$$

Assume 1 m³ volume of sample

Total weight of sample = 1900 kg

$$W_s + W_w = 1900 \text{ kg}$$

$$W_s + 0.19W_s = 1900 \text{ kg}$$

$$W_s = 1596.6 \text{ kg}$$

$$\text{Thus, } W_w = 0.19W_s$$

$$= 0.19 \times 1596.6 \text{ kg} = 303.4 \text{ kg}$$

Calculate volume of solids

From NCEES PE handbook page 2, mass of 1 cubic meter water = 1000 kg

$$\gamma_w = 1000 \text{ kg/m}^3$$

From page 134,

$$V_s = W_s / (G \cdot \gamma_w)$$

$$V_s = 1596.6 \text{ kg} / (2.7 \times 1000 \text{ kg/m}^3)$$

$$V_s = 0.591 \text{ m}^3$$

$$V_w = W_w / \gamma_w$$

$$V_w = 303.4 \text{ kg} / 1000 \text{ kg/m}^3$$

$$V_w = 0.3034 \text{ m}^3$$

Thus volume of air in 1 m³ sample

$$V_a = 1 \text{ m}^3 - 0.591 \text{ m}^3 - 0.3034 \text{ m}^3$$

$$V_a = 0.106 \text{ m}^3$$

Total volume of voids = $V_a + V_w$

$$V_v = 0.106 \text{ m}^3 + 0.3034 \text{ m}^3 = 0.409 \text{ m}^3$$

Degree of saturation = V_w / V_v (Ref NCEES PE handbook page 135)

$$= 0.3034 \text{ m}^3 / 0.409 \text{ m}^3$$

$$= 0.742 \approx \mathbf{75\%}$$
 Correct answer is option C

43. Problem

Correct Answer: (C) 4032 psi

Explanation:

Ref to ACI 318-14, page 478 Section 26.12.1.1,

Strength test shall be average of at least three 4X8 in cylinders

Area of samples

$$A = \{\pi * (4 \text{ in})^2\} / 4$$

$$A = 12.566 \text{ in}^2$$

$$\text{Sample 1 Strength} = 50 \text{ kips} / 12.566 \text{ in}^2 = 3.979 \text{ ksi} = 3979 \text{ psi}$$

$$\text{Sample 2 Strength} = 45 \text{ kips} / 12.566 \text{ in}^2 = 3.581 \text{ ksi} = 3581 \text{ psi}$$

$$\text{Sample 3 Strength} = 57 \text{ kips} / 12.566 \text{ in}^2 = 4.536 \text{ ksi} = 4536 \text{ psi}$$

$$\text{Average strength} = (3979 \text{ psi} + 3581 \text{ psi} + 4536 \text{ psi}) / 3$$

$$\text{Average strength} = \mathbf{4032 \text{ psi}}$$

Correct answer is option C.

44. Problem

Correct Answer: (C) 4

Explanation:

Effective depth, d = total depth - effective cover

$$d = 18 \text{ in} - 2 \text{ in} = 16 \text{ in}$$

$$B = 12 \text{ in}$$

Ref to ACI 318-14, page 137 section 9.6.1.2,

Minimum area of rebar should be greater of

$$\begin{aligned} \text{A) } & (3 \sqrt{f'_c} / f_y) * b * d \\ & = (3 \sqrt{3500} / 60000) * 12 \text{ in} * 16 \text{ in} \\ & = 0.57 \text{ in}^2 \end{aligned}$$

$$\begin{aligned} \text{B) } & 200 / f_y * b * d \\ & = 200 / 60000 * 12 \text{ in} * 16 \text{ in} \\ & = 0.64 \text{ in}^2 \end{aligned}$$

Thus minimum area of steel = 0.64 in^2

From NCEES PE handbook, page 260 section 4.3.1

Area of #4 bars = 0.2 in^2

No of bars required = $0.64 / 0.2 = 3.2$,

Hence **4** bars is required and the correct answer is Option C.

45. Problem

Correct Answer: (D) 47

Explanation:

Ref to ACI 318-14 page 72, Table 6.6.3.1.1 (a)

Moment of inertia for Columns = $0.7 I_g$

Moment of inertia of beams = $0.35 I_g$

For beams,

$$I_{\text{beam}} = 0.35 * 1/12 * 12 \text{ in} * (24 \text{ in})^3$$

$$I_{\text{beam}} = 0.35 * 13824 \text{ in}^4 = 4838.4 \text{ in}^4$$

For columns,

$$I_{\text{column}} = 0.7 * 1/12 * 14 \text{ in} * (14 \text{ in})^3$$

$$I_{\text{column}} = 0.7 * 3201.33 \text{ in}^4 = 2240.9 \text{ in}^4$$

For the same grade of concrete, the modulus of elasticity is the same.

From ACI 318-14 page 65, Fig R6.2.5,

For top joint

$$\Psi_a = (I_{\text{column}}/15\text{ft}) / (2 * I_{\text{beam}}/12.5\text{ft})$$

$$\Psi_a = [2240.9/(15*12)] / [2*4838.4/(12.5*12)]$$

$$\Psi_a = 0.193$$

For bottom joint

$$\Psi_b = (2 * I_{\text{column}}/15\text{ft}) / (2 * I_{\text{beam}}/12.5\text{ft})$$

$$\Psi_b = [2*2240.9/(15*12)] / [2*4838.4/(12.5*12)]$$

$$\Psi_b = 0.386$$

For sway frames, from ACI 318-14 page 65, Fig

R6.2.5, $k = 1.1$

Calculate radius of gyration for rectangular columns

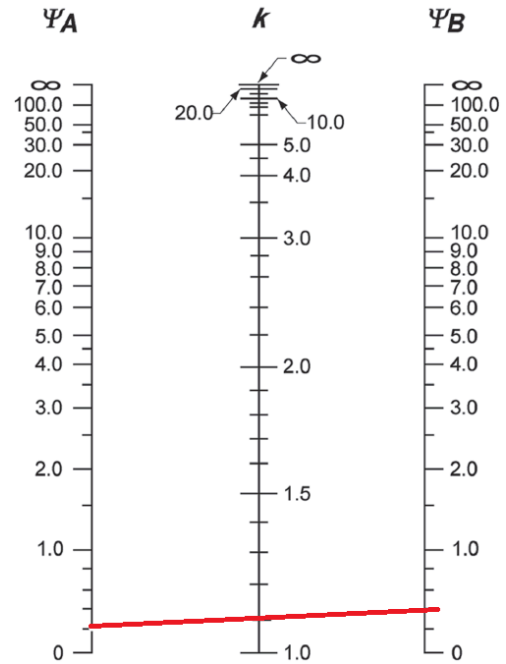
from page 64, section 6.2.5.1 (b)

$$r = 0.3 * 14 \text{ in} = 4.2 \text{ in}$$

$$\text{Slenderness} = kL/r$$

$$= 1.1 * 15 * 12 \text{ in} / 4.2 \text{ in} = 47.14$$

Correct answer is Option D, **47**.



46. Problem

Correct Answer: (A) 56 ksi and 33 ksi

Explanation:

Ref to AISC steel manual 15th edition page 16.1-129, AISC 360-16 Table J3.2

Nominal tensile strength = $0.75 F_u$

$$= 0.75 * 150 \text{ ksi} = 112.5 \text{ ksi}$$

Nominal shear strength = $0.45 F_u$

$$= 0.45 * 150 \text{ ksi} = 67.5 \text{ ksi}$$

From page 16.1-131, section J3.6 Strength reduction factor for allowable stress design = 2

$$\text{Allowable tensile strength} = 112.5 \text{ ksi} / 2 = 56.25 \text{ ksi}$$

$$\text{Allowable shear strength} = 67.5 \text{ ksi} / 2 = 33.75 \text{ ksi}$$

Thus correct answer is option A, **56 ksi and 33 ksi**

47. Problem

Correct Answer: (D) III only

Explanation:

Per AISC Steel manual 15th edition, page 16.1-7, AISC 306-16 section A3, 1c and 1d, Rolled heavy shapes and built up members with flanges and plate thickness exceeding 2 inches need Charpy V notch impact tests per ASTM A673. Hence the correct answer is option D, III only, which has a flange of 3 in.

48. Problem

Correct Answer: (B) 5.34

Explanation:

Ref to AISC Steel manual 15th edition, AISC 360-16, page 16.1-71 section G2.1(2) for web plate shear buckling coefficient k_v

Clear distance between the flanges

$$h = 15 \text{ in} - 2 \text{ in} - 2 \text{ in}$$

$$h = 11 \text{ in}$$

$$a = 40 \text{ in}$$

$$a/h = 40/11$$

$$a/h = 3.63 > 3$$

$$\text{Thus, } k_v = 5.34$$

Correct answer is option B.

49. Problem

Correct Answer: (B) 117 lbs

Explanation:

Ref to NDS 2018 page 111 Table 12N

For DFR with side member thickness 1.5 in and 16d common nails,

Nail diameter, $D = 0.162 \text{ in}$

Reference $Z = 141 \text{ lbs}$

Calculate penetration of nail

$$L/3 = 3.5 \text{ in} / 3$$

$$L/3 = 1.17 \text{ in}$$

$$\text{Penetration} = L \cdot \cos 30^\circ - L/3$$

$$= 3.5 \cdot \cos 30^\circ - 1.17 \text{ in}$$

$$= 1.86 \text{ in}$$

Check $10 \cdot D = 10 \cdot 0.162 = 1.62 \text{ in}$. Hence penetration is greater than $10D$,

No adjustment required for penetration

From NDS 2018 page 67, for dry conditions and 100°F , $C_m = C_t = 1$

From page 11, Table 2.3.2, for live loads, $C_d = 1$

From page 91, section 12.5.4, $C_{tn} = 0.83$

From page 89, section 12.5.1, $C_\Delta = 1$ for $D < 0.25 \text{ in}$

For this connection geometry, C_g and C_{di} are both equal to 1

From page 66, Table 11.3.1,

$$Z' = Z \cdot C_d \cdot C_m \cdot C_t \cdot C_g \cdot C_\Delta \cdot C_{eg} \cdot C_{di} \cdot C_{tn}$$

$$Z' = 141 \text{ lbs} \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 0.83$$

$$Z' = 117.03 \text{ lbs}$$

Correct answer is Option B, **117 lbs**.

50. Problem

Correct Answer: (C) 195 psi

Explanation:

Ref to TMS 402/602-16, page C-45 Section 5.1.3.1,

Vertical loads are dispersed at 2V:1H on both sides of the bearing plate.

At mid height i.e. 3ft, the load dispersion on each side of the bearing plate

$$= 3\text{ft}/2 = 1.5 \text{ ft} = 18 \text{ in}$$

Thus total effective length of dispersion

$$= 1.5 \text{ ft} + 5 \text{ in bearing plate} + 1.5 \text{ ft}$$

$$= 41 \text{ in}$$

Since mortar is not provided at the webs, thickness of masonry resisting the loads are only the two faces of the CMU unit and not webs.

$$\text{Effective thickness} = 2 \cdot 1.25 \text{ in} = 2.5 \text{ in}$$

$$\text{Axial stress due to load} = P/A$$

$$= 20 \text{ kips} / (41 \text{ in} \cdot 2.5 \text{ in})$$

$$= 0.195 \text{ ksi}$$

$$= \mathbf{195 \text{ psi}}$$

Correct answer is option C.

51. Problem

Correct Answer: (D)

Explanation:

Marking on rebar S denotes conformance to A615 carbon steel specs and W denotes conformance to A706 low alloy steel specs. Hence both Options A and B are not true. A706 rebars have lesser carbon content than A615 carbon steel. A706 has higher ductility and weldability. Option D is the correct answer.

52. Problem

Correct Answer: (C) I, II and III

Explanation:

The number of blows from an SPT test can be correlated to different soil properties like shear modulus, unconfined compressive strength, angle of internal friction, relative density and consistency, etc. Hence the correct answer is Option C.

53. Problem

Correct Answer: (B) W18X50

Explanation:

Here max unbraced length due to bracing at mid point = $25\text{ft}/2$

$L_b = 12.5\text{ ft}$

Max moment due to uniformly distributed = $w \cdot L^2 / 8$ (Ref. NCEES PE handbook page 236)

$M = 3\text{ kip/ft} \cdot (25\text{ ft})^2 / 8 = 234.37\text{ kip}\cdot\text{ft}$

From AISC Steel Manual 15th Ed, page 3-26, Table 3-2, M_p/Ω for W18x40 = $196\text{ kip}\cdot\text{ft}$ which is less than required, so it doesn't satisfy the requirement.

From AISC Steel Manual 15th Ed, page 3-18, Table 1, for this case, $C_b = 1.3$

From page 3-25, Table 3-2, for W18X50,

$M_p/\Omega = 252\text{ kip}\cdot\text{ft}$, $L_p = 5.83\text{ ft}$, $L_r = 16.9\text{ ft}$, $BF/\Omega = 8.76\text{ kips}$

From AISC steel manual page 3-10, when $L_p < L_b < L_r$

From equation 3-4b

$M_n/\Omega = C_b \cdot [M_p/\Omega - BF/\Omega \cdot (L_b - L_p)]$

$= 1.3 \cdot (252\text{ kip}\cdot\text{ft} - 8.76\text{ kip} \cdot (12.5\text{ ft} - 5.83\text{ft}))$

$= 251.6\text{ kip}\cdot\text{ft}$ (Check this should be less than M_p/Ω , OK here)

Hence the lightest W18 section that satisfies flexural strength requirement is **W18X50**.

Correct answer is Option B.

54. Problem

Correct Answer: (C) S8X23

Explanation:

Ref to AISC steel manual 15th ed., page 3-77, Table 3-7

The only section that satisfies the shear requirement is **S8X23**.

With $V_n/\Omega = 50.8$ kips,

Options A and B have allowable shear strength less than 46 kips.

Hence the most economical and correct answer is Option C.

55. Problem

Correct Answer: (B) $b = 7.625$ in , $d_v = 24$ in

Explanation:

Ref to TMS 402/602-16 page C-118, section 8.3.5.1.2 (b)

Max allowable shear stress in masonry for $M/Vd_v > 1$

$\gamma_g = 1$ for fully grouted section

$$F_v = 2[\sqrt{f'_m}] * \gamma_g$$

$$F_v = 2[\sqrt{1500}] * 1$$

$$F_v = 77.46 \text{ psi}$$

Shear stress acting on net shear area

$$F_v = V / (b * d_v)$$

$$77.46 \text{ psi} = 13000 \text{ lbs} / (7.625 \text{ in} * d_v)$$

$$d_v = 13000 \text{ lbs} / (77.46 \text{ psi} * 7.625 \text{ in})$$

$$d_v = 22.01 \text{ in}$$

Hence **$b = 7.625$ in , $d_v = 24$ in** is adequate and Option B is the answer.

56. Problem

Correct Answer: (C) 0.43 ksi (compressive)

Explanation:

Area of total 10 strands, $A_s = 10 * 0.153 \text{ in}^2 = 1.53 \text{ in}^2$

Total prestressing force, $P = 160 \text{ ksi} * 1.53 \text{ in}^2 = 244.8 \text{ kips}$

Area of section, $A = 40 \text{ in} * 24 \text{ in}$

$A = 960 \text{ in}^2$

Section modulus, $S = b*d^2/6$

$S = 24 \text{ in} * 40 \text{ in} * 40 \text{ in} / 6 = 6400 \text{ in}^3$

Eccentricity of prestressing strands from center of section

$e = 40 \text{ in} / 2 - 4 \text{ in} - 2 \text{ in} / 2 = 15 \text{ in}$

Bending stress due to prestress = $P*e/S$

$= 244.8 \text{ kips} * 15 \text{ in} / 6400 \text{ in}^3 = 0.574 \text{ ksi}$ (tension at top)

Compressive stress = P/A

$= 244.8 \text{ kips} / 960 \text{ in}^2 = 0.255 \text{ ksi}$ (compression at top)

Due to self weight, stress = 0.75 ksi (compression at top)

Total top fiber stress = 0.75 ksi + 0.255 ksi - 0.574 ksi = **0.43 ksi (compressive)**

Correct answer is Option C.

57. Problem

Correct Answer: (A) 360 plf

Explanation:

From AWC- SDPWS 2015, page 21, Table 4.2A,

For Structural I grade sheathing with thickness 3/8 inch and 8d nails @ 4in spacing, for seismic load, nominal shear strength assuming 2 in width of nailed face at boundaries,

$V_s = 720 \text{ plf}$

No adjustments are required for Southern Pine blocking.

From AWC- SDPWS 2015, page 16, section 4.2.3, ASD reduction factor = 2.0

Hence allowable diaphragm shear = $720 \text{ plf} / 2 = 360 \text{ plf}$

Correct answer is Option A.

58. Problem

Correct Answer: (C) 83 kips

Explanation:

Effective depth of slab, $d = 8 \text{ in} - 1.5 \text{ in} = 6.5 \text{ in}$

From ACI 318-14, page 341 Table 21.2.1,
strength reduction factor for shear, $\Phi = 0.75$

From ACI 318-14, page 362, Section 22.6.4.1,

The critical perimeter for edge column is

$$b_o = 2*(c_1 + d/2) + (c_2 + d)$$

$$b_o = 2*(12 \text{ in} + 6.5 \text{ in}/2) + (18 \text{ in} + 6.5 \text{ in})$$

$$b_o = 55 \text{ in}$$

Calculate two way shear strength from ACI 318-14, page 364, section 22.6.5

From page 364, Table 22.6.5.2

$$\beta = 18 \text{ in}/12 \text{ in} = 1.5$$

From page 365, section 22.6.5.3, $\alpha_s = 30$ for edge column

Assume normal weight concrete and $\lambda = 1$

v_c is the least of

$$\text{a) } 4\lambda(\sqrt{f'_c}) = 4*1*\sqrt{6000} = 309.8 \text{ psi}$$

$$\text{b) } (2+4/\beta)*\lambda*\sqrt{f'_c} = (2+4/1.5)*1*\sqrt{6000} = 361.48 \text{ psi}$$

$$\text{c) } (2+\alpha_s*d/b_o)*\lambda*\sqrt{f'_c} = (2+30*6.5 \text{ in}/55 \text{ in})*1*\sqrt{6000} = 429.5 \text{ psi}$$

$$v_c = 309.8 \text{ psi}$$

Shear strength

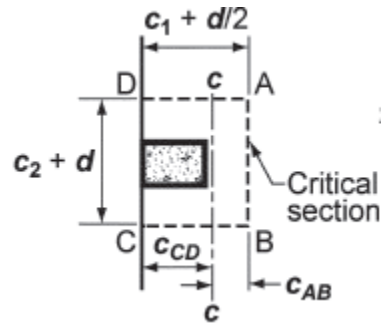
$$V = \Phi*v_c*b_o*d$$

$$V = 0.75*309.8 \text{ psi}*55 \text{ in}*6.5 \text{ in}$$

$$V = 83065.13 \text{ lbs}$$

$$V \approx \mathbf{83 \text{ kips}}$$

Correct answer is option C.



59. Problem

Correct Answer: (A) HSS 3 X 3 X 3/16

Explanation:

From AISC steel manual 15th ed., page 16.1-570, table C-A-7.1,

Effective length factor for pinned- fixed column, $k = 0.8$

Effective length

$$k * L = 0.8 * 13.5 \text{ ft}$$

$$k * L = 10.8 \text{ ft}$$

From steel manual page 4-82, table 4-4, for $Kl = 11 \text{ ft}$,

HSS 3 X 3 X 3/16 can withstand compression of 21.2 kips

Hence it is the lightest section that satisfies the requirement.

Correct answer is Option A.

60. Problem

Correct Answer: (D) 10.5 ksi

Explanation:

From AISC steel manual 15th ed. Page 16.1-140 and 16.1-141, section J8

$$\Omega_c = 2.31$$

Area of bearing plate

$$A_1 = 15 \text{ in} \times 15 \text{ in} = 225 \text{ in}^2$$

Area of frustum

$$A_2 = 18 \text{ in} \times 18 \text{ in} = 324 \text{ in}^2$$

$$\text{Check } A_2 < 4 * A_1 \quad 324 \text{ in}^2 < 4 * 225 \text{ in}^2 \quad \text{O.K.}$$

From eq. J8-2

$$P_p = 0.85 f'_c * A_1 * \sqrt{A_2/A_1}$$

$$1000 \text{ kips} = 0.85 * f'_c * 225 \text{ in}^2 * \sqrt{324 \text{ in}^2 / 225 \text{ in}^2}$$

$$f'_c = 1000 \text{ kips} / 229.5 \text{ in}^2$$

$$f'_c = 4.357 \text{ ksi}$$

$$\text{Using ASD factor } \Omega_c * f'_c = 2.31 * 4.357 \text{ ksi} = 10.1 \text{ ksi}$$

Hence the correct answer is Option D, **10.5 ksi**.

61. Problem

Correct Answer: (B) 38

Explanation:

From AISC Steel Manual 15th ed. page 1-26 and page 1-27, table 1-1

For W12X120

$$r_x = 5.51 \text{ in}$$

$$r_y = 3.13 \text{ in}$$

From steel manual page 16.1-570, table C-A-7.1

For pin fixed condition $K = 0.8$

For pin pin condition $K = 1.0$

For weaker axis, $L_y = 10\text{ft}$ with $K = 1$ (pin-pin for top half) governing

$$K \cdot L_y / r_y$$

$$= 1 \cdot (10 \cdot 12\text{in}) / 3.13\text{in} = 38.33$$

For stronger axis, $L_x = 20 \text{ ft}$ with $K = 0.8$ for fixed-pinned

$$K \cdot L_x / r_x$$

$$= 0.8 \cdot (20 \cdot 12\text{in}) / 5.51\text{in} = 34.8$$

Hence weaker axis governs and the correct answer is Option B, **38**.

62. Problem

Correct Answer: (A) 610 plf

Explanation:

From AWC- SDPWS 2015, page 35, Table 4.3A,

For Structural I panel, 3/8 in thick, 8d nails with spacing 2 in,

Nominal seismic shear strength = 1220 plf

No adjustment for southern pine framing is required.

From AWC- SDPWS 2015, page 25, section 4.3.3, ASD reduction factor = 2.0

$$\text{Allowable shear} = 1220 \text{ plf} / 2 = \mathbf{610 \text{ plf}}$$

Correct answer is Option A.

63. Problem

Correct Answer: (A) 3.5 in

Explanation:

When it is assumed that entire shear is taken by reinforcement only, $V_c = 0$

$$d = 18 \text{ in} - 3 \text{ in} = 15 \text{ in}$$

From ACI 318-14, page 341, Table 21.2.1

For shear, $\Phi = 0.75$

Ref to ACI 318-14, page 507, Appendix A for bar size areas

For #4 Ties (two legs of the tie are engaged),

$$A_v = 2 \times 0.2 \text{ in}^2 = 0.4 \text{ in}^2$$

From ACI 318-14, page 359, equation 22.5.10.5.3

$$V_s = (A_v \times f_y \times d) / s$$

$$\text{Required strength } V_s = 75 \text{ kips} / \Phi = 75 \text{ kips} / 0.75 = 100 \text{ kips}$$

$$s = (A_v \times f_y \times d) / 100 \text{ kips}$$

$$s = (0.4 \text{ in}^2 \times 60 \text{ ksi} \times 15 \text{ in}) / 100 \text{ kips}$$

$s = 3.6 \text{ in}$, spacing should be less than this.

Hence the correct answer is Option A, **3.5 in**.

64. Problem

Correct Answer: (D) Enhance lateral stiffness and resistance while the stiffer wall takes the majority of the lateral load.

Explanation:

Moment frames are combined with shear walls to enhance the overall strength and ductility of the shear wall only systems. The walls are typically much stiffer than the framing elements and carry the majority of the load. Hence Option D is the correct answer.

65. Problem

Correct Answer: (A) 263 kips

Explanation:

For HSS14.000X0.5, from AISC Steel manual 15th Ed, page 1-101 Table 1-13,

Area, $A_s = 19.8 \text{ in}^2$

$D/t = 30.1$

Given, $F_y = 46 \text{ ksi}$

$E_s = 29000 \text{ ksi}$

$f'_c = 4 \text{ ksi}$

From AISC Steel manual 15th edition, page 16.1-89, Table I1.1a

Limiting D/t for compact section $= 0.15 * E / F_y$

$= 0.15 * 29000 \text{ ksi} / 46 \text{ ksi} = 94.5$, i.e. section is compact

From AISC Steel manual 15th edition, page 16.1-91 section 1b

Strength reduction factor for axial compression of composites, $\Omega = 2$

Area of concrete, $A_c = (\pi/4) * D^2 - A_s = (\pi/4) * 14 \text{ in} * 14 \text{ in} - 19.8 \text{ in}^2$

$A_c = 134.1 \text{ in}^2$

From page 16.1-93 equation I2-9(a) and (b), for compact sections,

$P_{no} = P_p$

$P_p = F_y * A_s + C_2 * f'_c * (A_c + A_{sr} * E_s / E_c)$ (Area of rebar, $A_{sr} = 0$)

$P_p = 46 \text{ ksi} * 19.8 \text{ in}^2 + 0.95 * 4 \text{ ksi} * (134.1 \text{ in}^2)$

$P_p = 1420.4 \text{ kips} = P_{no}$

Given, elastic critical buckling strength, $P_e = 600 \text{ kips}$

$P_{no} / P_e = 1420.4 \text{ kips} / 600 \text{ kips} = 2.367$

From page 16.1-91 equation I2-3

When $P_{no} / P_e > 2.25$

$P_n = 0.877 * P_e$

$P_n = 0.877 * 600 \text{ kips} = 526.2 \text{ kips}$

Allowable strength

$P_n / \Omega = 526.2 \text{ kips} / 2 \approx \mathbf{263 \text{ kips}}$

Correct answer is Option A.

66. Problem

Correct Answer: (B) 450 kips

Explanation:

From AISC steel manual 15th ed., page 3-14, Fig. 3-3, the distance between top of steel and location of effective concrete flange force is Y_2

$$Y_2 = 3.5 \text{ in}$$

$$\text{Total service level moment} = 150 \text{ kip}\cdot\text{ft} + 290 \text{ kip}\cdot\text{ft} = 440 \text{ kip}\cdot\text{ft}$$

From AISC steel manual 15th ed., page 3-174, Table 3-19

For $Y_2 = 3.5 \text{ in}$ and W18X55 with allowable moment 441 kip·ft,

$$\Sigma Q_n = 454 \text{ kips}$$

Closest answer is Option B, **450 kips**.

67. Problem

Correct Answer: (C) Four #6 bars

Explanation:

Per ACI 318-14 provisions for special moment frames, page 277, Section 18.6.3.2

Positive moment strength at joint face shall be at least one-half the negative moment strength provided at that face of the joint.

From ACI 318-14, page 507, appendix A

$$\text{Area of \#8 bars} = 0.79 \text{ in}^2$$

$$\text{Total area of negative reinforcement} = 4 * 0.79 \text{ in}^2 = 3.16 \text{ in}^2$$

$$\text{Required area of positive reinforcement} = 0.5 * 3.16 \text{ in}^2 = 1.58 \text{ in}^2$$

$$\text{Area of six \#4 bars} = 6 * 0.2 \text{ in}^2 = 1.2 \text{ in}^2, \text{ Not enough}$$

$$\text{Area of four \#5 bars} = 4 * 0.31 \text{ in}^2 = 1.24 \text{ in}^2, \text{ Not enough}$$

$$\text{Area of four \#6 bars} = 4 * 0.44 \text{ in}^2 = 1.76 \text{ in}^2, \text{ O.K.}$$

Hence correct answer is Option C, **Four #6 bars**.

68. Problem

Correct Answer: (C) Min 3/16 in, Max 7/16 in

Explanation:

From AISC steel manual 15th ed. , page 16.1-119, Table J2.4, minimum size of fillet welds for joining plates 0.5 in thick is 3/16 in.

From AISC steel manual 15th ed. , page 16.1-119, along edges of material 1/4 in. or more in thickness; weld size should not be greater than the thickness of the material minus 1/16 in
 $= 1/2 \text{ in} - 1/16 \text{ in}$

$= 7/16 \text{ in}$

Hence the correct answer is Option C , **Min 3/16 in, Max 7/16 in.**

69. Problem

Correct Answer: (C) L 5 X 5 X 1/2

Explanation:

Since assumed failure is gross section failure, net area, bolt bearing etc. does not have to be checked.

From AISC steel manual 15th Ed. page 5-15 and page 5-16, Table 5-2

Tension capacity of L 5 X 3 X 1/2 = 80.8 kips

Tension capacity of L 5 X 3 1/2 X 1/2 = 86.2 kips

Tension capacity of L 5 X 5 X 1/2 = 103 kips, Meets requirement

Hence Correct answer is Option C.

70. Problem

Correct Answer: (A) 137 kips

Explanation:

The governing failure mode is single shear of bolts and bolt bearing/tearout.

From AISC steel manual, page 7-27, Table 7-4

For 3 inch spacing, standard holes with $F_u = 58 \text{ ksi}$, for 7/8 inch diameter bolts

Allowable bolt bearing = 60.9 kip/in

For 1/4 inch plate, Allowable bolt bearing/tearout = 60.9 kip/in * 1/4 in = 15.23 kips

From AISC steel manual, page 7-22, Table 7-1, For Group A 7/8 inch diameter bolts

For single shear, threads not included in shear plane,

V allowable = 20.4 kips

Thus bearing/tearout governs

Total number of bolts = 9

Allowable strength of connection = $9 * 15.23 \text{ kips} \approx \mathbf{137 \text{ kips}}$

Correct answer is Option A.

71. Problem

Correct Answer: (B)

Explanation:

Slip critical connections have lesser shear strength than bearing type connections for the same bolts. Slip critical connections do not increase the bolt hole tolerance or eliminate the need for bearing/tearout checks. They are designed to prevent bolt slip under service loads, ensuring stiffness and minimizing movements, especially in fatigue-sensitive or dynamically loaded structures. Hence Option B is the correct answer.

72. Problem

Correct Answer: (B) **1.2 kips**

Explanation:

The maximum force that can be applied to the angle is governed by the flexural yielding of the upright leg at the angle bend due to the applied load 'P'.

Thickness of leg, $t = 0.5 \text{ in}$

Width, $b = 4 \text{ in}$

Plastic section modulus,

$$Z = b * t^2 / 4 = 4 \text{ in} * (0.5 \text{ in})^2 / 4 = 0.25 \text{ in}^3$$

Elastic section modulus

$$S = b * t^2 / 6 = 4 \text{ in} * (0.5 \text{ in})^2 / 6 = 0.167 \text{ in}^3$$

$$1.6 * S > Z,$$

Hence allowable flexural strength

$$= F_y * Z / \Omega \quad (\Omega = 1.67 \text{ from AISC steel manual 15th ed page 16.1-46, section F1})$$

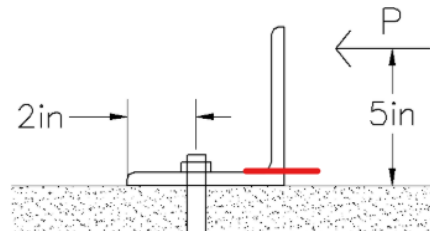
$$= 36 \text{ ksi} * 0.25 \text{ in}^3 / 1.67 = 5.39 \text{ kip*in}$$

$$\text{Moment arm due to } P = (5 \text{ in} - 0.5 \text{ in}) = 4.5 \text{ in}$$

$$\text{Allowable moment} = P * \text{moment arm}$$

$$P = 5.39 \text{ kip*in} / 4.5 \text{ in} = \mathbf{1.2 \text{ kips}}$$

Correct answer is Option B.



73. Problem

Correct Answer: (B) 2 in

Explanation:

Total load = Dead + Live = 20 kips + 30 kips = 50 kips

Size of 3/8 inch weld in 16th of an inch, $3/8 \text{ in} = 6/16 \text{ in}$. Hence $D = 6$

From AISC steel manual 15th ed. Page 8-8, eqn. 8-2b, $R_n/\Omega = 0.928 * D * l$

From AISC steel manual 15th ed. Page 16.1-124, eqn. J2-6a and eqn. J2-6b,

For transverse and longitudinal welds, the nominal resistance can be taken as greater of

(i) $R_n = R_{nwl} + R_{nwt}$

i.e. $R_n/\Omega = (R_{nwl}/\Omega) + (R_{nwt}/\Omega)$

(i) $R_n = 0.85 R_{nwl} + 1.5 * R_{nwt}$

i.e. $R_n/\Omega = 0.85 (R_{nwl}/\Omega) + 1.5 * (R_{nwt}/\Omega)$

Using strength reduction factor, Ω

Try, $L_w = 1.5 \text{ in}$

Total longitudinal length = $2 * 1.5 \text{ in} = 3 \text{ in}$

Total transverse length = 4 in

Allowable strength (i) = $0.928 * 6 * 3 + 0.928 * 6 * 4 = 38.97 \text{ kips}$

Allowable strength (ii) = $0.85 * 0.928 * 6 * 3 + 1.5 * 0.928 * 6 * 4 = 47.6 \text{ kips}$ (not enough)

Try, $L_w = 2 \text{ in}$

Total longitudinal length = $2 * 2 \text{ in} = 4 \text{ in}$

Total transverse length = 4 in

Allowable strength (i) = $0.928 * 6 * 4 + 0.928 * 6 * 4 = 44.5 \text{ kips}$

Allowable strength (ii) = $0.85 * 0.928 * 6 * 4 + 1.5 * 0.928 * 6 * 4 = 52.3 \text{ kips}$ (enough)

Hence the correct answer is Option B, **2 in**.

74. Problem

Correct Answer: (B) 6in

Explanation:

From ACI 318-14 page 507, Appendix A, diameter of #8 bar, $d_b = 1\text{ in}$

From ACI 318-14 page 142, section 9.7.3.3, from point of inflection, bars should extend at least by a length that is greater of 'd' or $12*d_b$

$$d = 21\text{ in} \quad 12*d_b = 12\text{ in}$$

Point of inflection is 2.5ft i.e 30 in. from the face of support.

Bars should be extended to $30\text{ in} - 21\text{ in} = 9\text{ in}$. away from support

Hence this clause does not govern extension of bars into the column

From ACI 318-14 page 143, section 9.7.3.8.2, at least one-fourth of the maximum positive moment reinforcement shall extend along the beam bottom into the support at least **6 in**.

Hence the correct answer is Option B, **6in**.

75. Problem

Correct Answer: (B) 10 in

Explanation:

From ACI 318-14 page 507, Appendix A, diameter of #6 bar, $d_b = 0.75\text{ in}$

From ACI 318-14, page 418, Table 25.4.3.2

Since the bar is hooked into a column that continues to the floors below, cover under the 90 degree extension is $> 2\text{ in}$, $\Psi_c = 0.7$

Since information about stirrups is not provided, $\Psi_r = 1.0$

From ACI 318-14, page 417, Section 25.4.3.1, L_{dh} should be greater of

a) $L_{dh} = (f_y * \Psi_e * \Psi_c * \Psi_r) / (50 * \lambda * \sqrt{f'_c}) * d_b$

$$L_{dh} = (60000 * 1 * 0.7 * 1) / (50 * 1 * \sqrt{4000}) * 0.75\text{ in} = 9.96\text{ in}$$

b) $8*d_b = 8 * 0.75\text{ in} = 6\text{ in}$

c) 6 in

Correct answer is Option B, **10 in**.

76. Problem

Correct Answer: (A) 9130 psf

Explanation:

Depth of footing $D_f = 5 \text{ ft}$

Depth of water table, $D_w = 2.5 \text{ ft} = 5\text{ft}/2 = D_f/2$

From NCEES PE handbook page 91, Bearing capacity factors,

For $\Phi = 30^\circ$, $N_q = 18.4$ $N_\gamma = 22.4$

When $D_w = D_f/2$, correction factor

$C_{wq} = 0.5$ $C_{w\gamma} = (1+0.5)/2 = 0.75$

Corrected bearing capacity factors

$N_q = 18.4 * C_{wq} = 18.4*0.75 = 13.8$

$N_\gamma = 22.4 * C_{w\gamma} = 22.4*0.5 = 11.2$

From NCEES PE handbook page 90, section 3.4.2

$q_{ult} = c * N_c + q * N_q + 0.5 * \gamma * B_f * N_\gamma$

$q_{ult} = 0 + 110 \text{ pcf} * 5\text{ft} * 13.8 + 0.5 * 110 \text{ pcf} * 2.5\text{ft} * 11.2 = \mathbf{9130 \text{ psf}}$

Correct answer is Option A.

77. Problem

Correct Answer: (B)

Explanation:

The assumptions stated by Options A, C and D are incorrect or applicable only in typical design scenarios.

The soil can be modelled as a series of independent linear springs with a subgrade modulus, which is also known as The Winkler model.

The best answer is Option B

78. Problem

Correct Answer: (C) 25 in

Explanation:

Factored shear, $V_u = 150$ kips

Total width of the section, $b_w = 6 \text{ ft} = 72 \text{ in}$

$f'_c = 4000$ psi

From ACI 318-14, page 341, Table 21.2.1, $\Phi = 0.75$ for shear

From ACI 318-14, page 354, equation 22.5.5.1

$$V_c = 2\lambda(\sqrt{f'_c}) * b_w * d$$

$$V_u = \Phi * V_c$$

$$V_u = 0.75 * 2 * (\sqrt{4000}) * 72 \text{ in} * d$$

$$d = 150 * 1000 \text{ lbs} / [0.75 * 2 * (\sqrt{4000}) * 72 \text{ in}] = 21.96 \text{ in}$$

$$\text{Depth of footing} = d + \text{effective cover} = 21.96 \text{ in} + 3 \text{ in} = 24.96 \text{ in}$$

Hence, the correct answer is Option C, **25 in**.

79. Problem

Correct Answer: (A)

Explanation:

Negative skin friction is the downward drag caused in the piles due to downward movement of surrounding soil due to settlement/consolidation.

Option B, C and D are all incorrect.

Option A is the answer.

80. Problem

Correct Answer: (D) 2091 lb*ft/ft

Explanation:

Active earth pressure coefficient, $K_a = 0.33$

Density of soil, $\gamma = 110$ pcf

Height of stem = 7ft- 1ft = 6ft

Total lateral load acting on the stem follows triangular distribution

Total load, $P = (1/2) * K_a * \gamma * H^2$

$P = (1/2) * 0.33 * 110 \text{ pcf} * (6 \text{ ft})^2$

$P = 653.4 \text{ lbs/ft}$

Moment arm = distance from stem base to centroid of triangle = $6\text{ft}/3 = 2\text{ft}$

Max moment in the stem = $P * \text{moment arm}$

$= 653.4 \text{ lbs/ft} * 2\text{ft}$

$= 1306.8 \text{ lb*ft/ft}$

From ACI 318-14, page 60, section 5.3.8, load factor for earth pressure = 1.6

Factored moment = $1.6 * 1306.8 \text{ lb ft/ft} \approx \mathbf{2091 \text{ lb*ft/ft}}$

Correct answer is option D.

